



Airease: Smart Exhaust Fan with Predictive Control

Mohit Sri Harsha Kajuluri
Chintala Yeswanth

This thesis is submitted to the Department of Mathematics and Natural Sciences at Blekinge Institute of Technology in partial fulfilment of the requirements for the degree of Bachelor of Science in Electrical Engineering. The thesis is equivalent to 10 weeks of full time studies.

Contact Information:

Author(s):

Mohit Sri Harsha Kajuluri

E-mail: mokj24@student.bth.se

Chintala Yeswanth

E-mail: yech24@student.bth.se

University advisor:

Irina Gertsoyich

Department of Mathematics and Natural Sciences

Department of Mathematics and Natural Sciences Internet : www.bth.se

Blekinge Institute of Technology

Phone : +46 455 38 50 00

SE-371 79 Karlskrona, Sweden

Fax : +46 455 38 50 57

Abstract

This thesis presents the design and implementation of a machine learning (ML)-based bathroom humidity control system aimed at energy efficiency and preventing mold growth. Conventional bathroom ventilation systems rely on manual controls or fixed humidity thresholds, which often result in excessive energy use or insufficient ventilation. The developed system integrates an ESP32 microcontroller with temperature, water flow, and humidity sensors to monitor environmental conditions in real time.

A lightweight ML model, trained using TinyML techniques, is deployed on the ESP32 to forecast short-term humidity levels based on live sensor data. Using the predicted humidity trends, the system proactively activates an exhaust fan via a relay module before humidity exceeds critical levels, thereby ensuring timely ventilation. Experimental evaluation demonstrates that the system effectively predicts humidity surges and responds with optimized fan control, offering a cost-efficient, scalable, and privacy-preserving solution for smart home ventilation.

Keywords: Humidity control, TinyML, ESP32, Smart ventilation, Predictive system

Acknowledgments

We extend our deepest gratitude to everyone who assisted us throughout the development of this thesis project. We are deeply thankful to our supervisor, Irina Gertsovich, for her valuable feedback, support, and continuous encouragement throughout the research and development process. Her advisory role was instrumental in shaping the way this work evolved. We would also like to thank Saleh Javadi for his support and for providing us with the necessary hardware resources, which greatly contributed to the practical implementation of our work. Our thanks also go to the departmental faculty members of the Department of Electrical Engineering at Blekinge Institute of Technology for providing a solid academic foundation and access to relevant resources and lab facilities. The unconditional support, encouragement, and understanding of our family and friends deserve special mention for their persistent encouragement during this effort. Finally, we appreciate the open-source websites and communities that provided technical advice, as well as open-source embedded system-related tools and resources for machine learning and sensor integration, all of which greatly assisted in the completion of this project.

Contents

Abstract	iii
Acknowledgments	v
List of Figures	viii
List of Tables	viii
1 Introduction	1
1.1 Background	1
1.2 Research Gap	2
1.3 Aim and Objectives	2
1.4 Research Question	2
1.5 Predictive Fan Control: Concept and Example	3
1.6 Scope of Thesis	3
1.7 Outline	3
1.8 Ethical, Societal and Sustainability Aspects	4
2 Related Work	5
3 Methodology	7
3.1 TinyML	7
3.1.1 Deployed Machine Learning Models	8
3.1.2 Summary	8
3.1.3 Mathematical Modeling of Humidity Accumulation	9
3.2 System Overview	10
3.2.1 System Schematic	10
3.3 Working Principle	10
3.3.1 Key Features & Logic	13
3.4 Component Selection and Justification	14
3.4.1 Sensor Configuration and Setup	16
3.4.2 Data Collection Conditions	16
3.5 Justification of Chosen Method	17
3.6 Validation and Testing	17
3.7 Validity and Reliability Considerations	18
3.8 Limitations and Self-Critique	19
3.9 Conclusion of Methodology	19

4	Results and Analysis	21
4.1	Training Data for the Model	21
4.1.1	Dataset Characteristics	23
4.1.2	Preprocessing and Labeling	23
4.1.3	Summary	24
4.1.4	Feature Correlation Analysis	24
4.1.5	3D-Printed Prototype Design and Bathroom Integration . . .	25
4.2	Model Performance	28
4.3	Gradient Boosting Time Series Prediction	29
4.3.1	Sample-Based Validation of Fan Control Logic	34
4.4	Limitations and Discussion	35
4.5	Conclusion of Analysis	35
5	Discussion	37
5.1	Revisiting Research Questions and Model Findings	37
5.2	Analysis of Prediction Quality	37
5.3	Embedded Inference and System Responsiveness	38
5.4	Sensor Fusion and Feature Importance	38
5.5	Energy Efficiency Assumption and Estimation	38
5.6	Limitations and Model Generalizability	39
5.7	Practical Implications and Ethical Reflections	39
5.8	Reconnecting with Research Objectives	40
5.9	Placing the Work in a Wider Perspective	40
5.10	Final Reflections	40
6	Conclusions and Future Work	41
6.1	Summary of Contributions	41
6.2	Future Work and Recommendations	41
6.3	Conclusion	42
	References	43
	References	43
A	Supplemental Information	47

List of Figures

3.1	System schematic of smart relative humidity control setup	10
3.2	Flowchart of the smart relative humidity prediction and fan control system	11
3.3	DS18B20 temperature sensor	14
3.4	YF-S201 water flow sensor	14
3.5	DHT22 humidity sensor	15
3.6	ESP32 microcontroller	15
3.7	Exhaust fan	16
3.8	Relay module used for exhaust fan control	16
4.1	Top view of the hardware prototype: ESP32, sensors, and relay module	22
4.2	Live testing of the prototype in a bathroom setup	22
4.3	relative humidity vs. Time showing shower and fan activation phases.	23
4.4	Heatmap showing Pearson correlation coefficients among humidity, flow rate, and water temperature,	25
4.6	Full system setup inside a shower stall, showing the waterproof prototype and exhaust fan for humidity ventilation.	28
4.7	Gradient Boosting Time Series Prediction showing actual and forecasted relative humidity (Test Set)	31
4.8	Actual vs. Predicted relative humidity with 45° reference line	31
4.9	Scatterplot showing flow rate vs. water temperature during showers. Scatter suggests phase-wise or user-level variability despite high overall correlation.	32
4.10	Histograms of humidity, flow rate, and water temperature.	33
4.11	Fan activation behavior based on measured and predicted relative humidity values during a sample shower cycle	34

List of Tables

3.1	Feature and its purposes	13
4.1	Model performance comparison	29

1.1 Background

Indoor air quality plays a vital role in achieving healthy and comfortable indoor conditions within a house. Among all the rooms within a residence, bathrooms are the most prone to uncontrolled humidity buildup due to frequent hot water usage through showering. This excess humidity, if not well managed, leads to the development of molds—a nagging and health-threatening problem in most residences worldwide. Studies by the U.S. Environmental Protection Agency (EPA) and Mayo Clinic highlight the seriousness of health risks that follow mold exposure, which can involve allergic responses, respiratory infections, asthma attacks, and chronic sinusitis, especially in children, the elderly, and immunocompromised individuals [EPA; Mayo Clinic]. Mold spores are carried in indoor air, aggravating previous conditions and triggering new ones through extended exposure [1–3].

From both health and economic perspectives, mold poses a serious challenge in indoor environments—especially in bathrooms with poor humidity control. According to the Mayo Clinic, WHO, and EPA, exposure to indoor mold can trigger asthma, allergies, and long-term respiratory complications, particularly affecting children and the elderly [4]. Structurally, mold leads to degradation of materials such as drywall and insulation, causes peeling paint, wall cracking, and structural weakening. In severe cases, full-scale renovations may be necessary, with reported costs ranging from approximately \$500 for surface-level treatments to over \$10,000 for comprehensive repairs [4]. These figures illustrate that mold is not only a hidden health hazard but also a significant financial burden in many households.

Therefore, mold prevention at its source excessive humidity is both a public health priority and an economic imperative. This thesis addresses these dual challenges by implementing predictive humidity control using embedded AI. By preventing mold formation in advance, the proposed system contributes directly to improved health outcomes and reduced maintenance costs, thereby supporting long-term housing sustainability.

Traditional bathroom ventilation systems typically operate via simple humidity-level sensors or manual switches. These reactive systems often respond too late, allowing moisture to accumulate and increasing the risk of mold buildup. Additionally, such systems contribute to energy inefficiency, either by remaining active when unnecessary or failing to turn on in time. Although modern fan modules offer automation, they lack learning capability and rely on static rules and pre-defined environmental conditions [5, 6].

Despite some advancements in IoT-based ventilation systems, most commercially available bathroom exhaust fans still rely on fixed timers or simple humidity threshold triggers, which often activate too late or run inefficiently. These reactive designs fail to account for evolving conditions such as shower patterns, water temperature, or occupancy. Prior work by A.Bind . [7] and the “IoT Smart Exhaust Fan” project [8] demonstrate systems with automated control but lack predictive or adaptive machine learning capabilities. This gap presents an opportunity to introduce a smarter, more responsive solution using embedded AI for real-time optimization.

In recent years, advances in embedded systems, Internet of Things (IoT) technologies, and Tiny Machine Learning (TinyML) have enabled more intelligent, real-time solutions. TinyML allows machine learning inference to be executed locally on low-power microcontrollers like the ESP32, eliminating the need for cloud infrastructure or continuous internet access [9,10]. This edge computing approach enables privacy-preserving and timely control in resource-constrained environments such as bathrooms.

In this thesis, we to bridge that gap by developing a privacy-preserving, low-cost, and predictive bathroom ventilation system. Using sensor data from water flow, temperature, and humidity sensors [11–13], we trained a TinyML model to forecast rising humidity levels and activate an exhaust fan accordingly. This anticipatory system improves comfort and energy efficiency while supporting public health by preventing mold growth before it becomes visible or structurally damaging. Our work highlights the potential of embedded AI to address real-world indoor air quality challenges, particularly in underexplored domestic environments such as bathrooms.

1.2 Research Gap

Present day bathroom exhaust fans are very good at minimizing indoor humidity levels but do so predominantly through pre-set timers or primitive threshold-level humidity sensors, rather than machine learning-based adaptive control. Consequently, such systems are incapable of predicting humidity fluctuations, optimizing the operation of the fan in accordance with usage patterns, or adjusting to environmental changes in real time. Despite the advancement of smart home technologies, machine learning-powered bathroom ventilation systems, particularly with on-device inference embedded, have been largely absent from research and industrial domains. This creates a shortage of intelligent, privacy-preserving, and energy-efficient humidity control methods for enclosed high-moisture areas.

1.3 Aim and Objectives

The main objective of this thesis is to design and implement a predictive ventilation control system for bathroom areas using embedded systems and TinyML in order to prevent mold and excessive humidity, reduce energy consumption, and enhance occupant comfort.

1.4 Research Question

1. What are the best ways to implement embedded AI with low-cost microcontrollers to predict and control bathroom humidity in real-time?

2. What is the most effective method for predicting humidity changes in a closed space ?
3. How can the system automatically control the exhaust fan to maintain optimal conditions?

1.5 Predictive Fan Control: Concept and Example

The core logic of this project is to give the bathroom exhaust fan the ability to improve. Instead of turning on only after the air becomes too humid, the system uses current sensor readings such as humidity, water temperature and flow rate to predict what the humidity will be in the next few minutes. If the predictions show that the humidity is about to cross a dangerous threshold, **the fan turns on in advance.**

For example, when some one starts taking a hot shower, the system detects the rising temperature and water flow. Even if the current humidity is still moderate, for instance 65%, the machine learning model may predict that it will rise to 75% shortly. In response, the fan activates early, helping to control humidity before it becomes a problem. This proactive approach improves comfort, prevents mold growth and reduces unnecessary fan usage, leading to better energy efficiency.

This predictive control makes the exhaust fans smarter than traditional systems, which only respond after conditions have already worsened.

1.6 Scope of Thesis

The main objective of this thesis is to design and test a predictive humidity control system for bathroom use in domestic settings using embedded AI techniques. The research focuses on this specific application to explore how sensor information, i.e., temperature and water supply, can be used with TinyML models running on resource-constrained microcontrollers like the ESP32.

This focus allows for extensive investigation of data acquisition, model training and implementation and dynamic control of an exhaust fan for maintaining optimal humidity. The work avoids large-scale smart home integration, other air conditioning systems and large-scale environmental variables outside the bathroom space. By narrowing the scope, the thesis is able to work towards building a pragmatic, energy efficient and scalable embedded solution.

The overarching research question underpinning this work is:

What are the best ways to implement TinyML with low-cost microcontrollers to predict and control bathroom humidity in real-time?

Other detailed research sub-questions and hypotheses about sensor performance, prediction accuracy, system latency and energy savings will be addressed in the Methodology chapter.

1.7 Outline

The research follows a specific structure to present the design process and implementation and testing of TinyML technology-based predictive bathroom humidity control systems.

Chapter 1: Introduction: The introduction provides background information about bathroom humidity control while explaining the research motivation and scope and objectives and discussing relevant ethical and societal and sustainability issues.

Chapter 2: Related Work: The chapter presents existing ventilation solutions and embedded system applications for environmental monitoring and machine learning model-based prediction control. This chapter explains the research gaps that this thesis aims to address.

Chapter 3: Methodology: The chapter explains hardware and software components and data collection procedures and machine learning model development and training and TinyML deployment on the ESP32 platform. The experimental setup and conditions receive detailed explanation in this section

Chapter 4: Results and Analysis: The chapter presents experimental results which include model performance data and system responsiveness and power efficiency measurements. The system performance evaluation under different operational conditions is presented in the data analysis section.

Chapter 5: Discussion: The chapter presents results discussion followed by an evaluation of embedded AI application strengths and limitations for humidity control. The chapter discusses practical applications and potential enhancement areas for the system.

Chapter 6: Conclusions and Future Work: The final section provides a summary of main contributions while addressing research questions and suggesting future directions to enhance and expand the system.

1.8 Ethical, Societal and Sustainability Aspects

The research investigates the design of an TinyML humidity bathroom control system through multiple ethical social and sustainability perspectives. The system operates autonomously to monitor environmental factors such as water usage and temperature without collecting personal data which protects user privacy and maintains their safety.

The system removes health dangers from mold development and excessive moisture while delivering better indoor air quality and decreased maintenance requirements which benefit users and public health. The system promotes sustainability through smart energy management because it operates ventilation systems only when necessary which reduces electricity usage and extends the lifespan of ventilation equipment.

The system operates with low-power hardware and affordable sensors which makes it possible to deploy it extensively at minimal environmental expense. The project promotes sustainable technology use and public health and environmental sustainability for future generations [1–3].

Understanding and improving indoor air quality—especially in moisture-prone areas like bathrooms—has been a focus of many recent research efforts. Various approaches have been proposed, including threshold-based ventilation, fuzzy logic control, cloud-based mold prediction, and machine learning applications. However, most existing systems tend to fall short when it comes to predicting changes before they occur, protecting user privacy, or operating efficiently in small, enclosed spaces. This thesis aims to bridge those gaps by using embedded machine learning (TinyML) to forecast humidity trends in advance and activate ventilation systems proactively.

Recent studies have shown that TinyML can be effectively integrated into low-power microcontrollers such as ESP32 and STM32 to perform real-time environmental monitoring and decision-making [14–16]. These works explored using machine learning on embedded systems for tasks like flow anomaly detection and leakage monitoring. The success of those implementations confirms the potential of TinyML for real-world environmental control. Building on this foundation, our project applies similar techniques to predict humidity changes and manage bathroom ventilation before moisture reaches harmful levels.

A notable study by Fadli Pradityo and Nico Surantha introduced a fuzzy logic-based control system for indoor air quality [17]. While effective for general indoor environments, their system lacked predictive capabilities and wasn’t designed with bathroom-specific conditions in mind. By contrast, our system uses embedded machine learning to anticipate humidity spikes, allowing the exhaust fan to react ahead of time and prevent mold buildup, rather than just responding after the fact.

Similarly, the smart fan system proposed by A. Bind and colleagues uses IoT technology to automatically adjust fan speed based on environmental readings [7]. While this improves responsiveness, their approach remains rule-based and reactive. Our solution advances this concept by incorporating TinyML models that can forecast future humidity levels, offering more timely and energy-efficient ventilation.

There are also studies that explore cloud-based machine learning models for predicting mold growth or food spoilage by analyzing humidity and temperature data over time [16]. While insightful, these models depend on a constant internet connection and suffer from latency, making them less suitable for real-time control. Our approach performs all computations locally on the ESP32 chip, ensuring immediate action without relying on cloud services.

The link between poor humidity control and mold growth is well documented. Bathrooms, with their frequent hot water usage and limited ventilation, are especially vulnerable. Studies from public health organizations such as the Mayo Clinic, EPA,

and WHO confirm that exposure to indoor mold can cause allergic reactions, respiratory problems, and long-term health complications—particularly in children, the elderly, and those with compromised immune systems. Mold also causes structural damage that can be expensive to repair. Despite these serious risks, most commercial ventilation systems still rely on simple threshold-based control. Our system directly addresses this gap by using AI to forecast humidity increases and activate ventilation early, helping prevent mold before it has a chance to form [5, 6, 10].

Research by Niranjana et al. and Wei et al. has explored the use of smart sensors in bathrooms for general automation and health monitoring [5, 6]. While these systems offer promising capabilities, they don’t include any mechanism for predictive humidity control. Our system extends their work by adding intelligent fan control based on forecasted moisture conditions, offering a more complete solution for maintaining indoor air quality.

AirNet and other similar efforts have developed machine learning models for predicting air quality in large spaces like offices and living rooms [10]. While their models are effective in broader environments, they are not designed for small, moisture-intensive settings like bathrooms. Our system is optimized for such spaces, using localized processing and low-power components to maintain performance without compromising privacy or efficiency.

Another smart bathroom project by Yu-Hsien Wei and colleagues demonstrates comprehensive monitoring and automation using a variety of sensors [6]. Although their system covers several aspects of smart home integration, it lacks the ability to anticipate changes in humidity. Our solution builds on this idea by integrating TinyML for proactive fan activation, which significantly enhances mold prevention.

In addition, time-series forecasting methods like ARIMA have been used to predict indoor humidity levels [18]. While statistically sound, these models often require more memory and computing power than typical microcontrollers can provide. By using a quantized Gradient Boosting model tailored for TinyML deployment, we are able to deliver fast and reliable forecasts using limited hardware resources.

Lastly, tutorials and open-source projects showcasing ESP32-based exhaust fans typically implement simple rule-based logic to control ventilation [8]. While they demonstrate how to get started with smart fan systems, they do not support learning or adaptive behavior. Our work significantly improves upon this by embedding AI-based decision-making into the system, resulting in smarter, faster, and more energy-conscious responses to changing bathroom conditions.

In conclusion, while the literature offers many promising directions—from fuzzy logic to cloud-based analytics—most existing systems either react too slowly, depend on external networks, or are not tailored to the unique challenges of bathrooms. Our proposed solution addresses these issues by offering a lightweight, embedded, and predictive system designed to keep bathrooms mold-free and energy efficient through intelligent, proactive ventilation control.

The main objective of this research is to develop an independent smart bathroom humidity control system that predicts humidity levels in real time and pre-controls an exhaust fan to provide comfortable and mold-free environments. The system integrates sensor data sensing, embedded TinyML-based real-time humidity prediction and fan control actuation to maximize indoor air quality and energy efficiency.

3.1 TinyML

TinyML, short for *Tiny Machine Learning*, refers to the process of executing machine learning (ML) models directly on small, low-power hardware platforms such as ESP32 or Arduino-based microcontrollers. Unlike traditional ML systems, which rely on high-performance computing platforms or cloud infrastructures, TinyML operates efficiently on devices with significantly limited resources. For example, the ESP32 microcontroller typically features around 500 KB of RAM and 4 MB of flash memory, which are sufficient for deploying compact models optimized for real-time inference.

In TinyML workflows, the model training phase is conducted on a computer using standard ML tools. Once trained, models are converted into a lightweight form suitable for deployment on embedded systems.

Advantages of TinyML: TinyML offers several practical advantages, particularly in embedded systems. It enables private and fully offline applications without requiring an internet connection. TinyML devices are designed for low power consumption, allowing them to operate for extended periods—months or even years—on battery power. Additionally, they support real-time processing, providing instant responses to sensor inputs without the latency associated with cloud-based systems. These characteristics make TinyML highly suitable for cost-efficient IoT and home automation projects.

Available TinyML Tools: A variety of tools have emerged to facilitate the development and deployment of TinyML models, including TensorFlow Lite for Microcontrollers (TFLM), Edge Impulse, and Arduino NanoML. This study employed TensorFlow Lite for Microcontrollers (TFLM) to deploy a quantized Gradient Boosting model onto the ESP32 platform. The model was encoded in FlatBuffer format and executed using TFLM's MicroInterpreter, enabling on-device inference with low memory usage and no reliance on cloud connectivity.

3.1.1 Deployed Machine Learning Models

In this project, supervised machine learning algorithms were employed to forecast humidity levels based on water flow rate, temperature, and humidity sensor data. Four models were evaluated: Linear Regression (LR), Random Forest (RF), Gradient Boosting (GB), and Support Vector Regression (SVR). LR established a simple linear relationship between inputs and output but lacked the capacity to model nonlinear patterns [19]. RF, an ensemble of decision trees, improved prediction accuracy by handling complex data distributions [20]. GB constructed a series of small trees where each subsequent tree corrected the errors of the previous, and its TinyML-compatible version (via microML or sklearn-porter) allowed efficient deployment on ESP32 with less than 20 KB memory usage [21]. SVR aimed to maximize margin tolerance but was less suitable for microcontrollers due to high computational demands [22].

Model training process : The process of training and deploying the predictive model involved structured steps aligned with TinyML development. The model inputs included water flow rate measured in litres per minute, temperature of water in degrees Celsius, and ambient humidity in percentage. The output was the forecasted humidity a few minutes ahead of the current reading.

The execution pipeline began with sensor data collection using an ESP32-based embedded system. The acquired data was cleaned and normalized, with all values scaled between 0 and 1 to ensure uniformity during training. Several machine learning models were developed and trained on a PC using Python and the scikit-learn library. After evaluating performance, the Gradient Boosting model was selected as the best performer.

This model was then exported and converted into TensorFlow Lite format. (TFLM) was utilized to deploy the quantized model on the ESP32 platform. Inference was executed using TFLM’s MicroInterpreter, which allowed efficient on-device predictions without cloud connectivity. The final quantized model used 8-bit integers and was successfully deployed, occupying less than 20 KB of memory, making it suitable for real-time embedded applications.

Dataset requirements : Sensor data was collected over three days in a real bathroom setting, resulting 6,000 samples. Each entry included humidity, water flow rate, and water temperature, capturing key environmental changes during shower use. Flow rates ranged from 0 to 15.33 L/min, water temperatures from 23.5°C to 38°C, and Relative humidity levels from 51.7% to 90.4%. This variability was sufficient to train a compact model for short-term humidity prediction within a single bathroom context.

3.1.2 Summary

TinyML enables the execution of ML models on microcontrollers, allowing for intelligent, responsive, and cost-effective edge devices. In this project, four machine learning models were evaluated, with Gradient Boosting exhibiting the highest performance for humidity prediction. Despite a relatively small dataset, the results were

reliable, and the final model was successfully deployed on the ESP32 platform.

3.1.3 Mathematical Modeling of Humidity Accumulation

To complement our machine learning approach, we developed a physics-informed, rule-based mathematical model that describes the relationship between water usage and relative humidity accumulation in a bathroom environment. The analytical model serves as a useful starting point to understand humidity accumulation based on water flow but it relies on several simplifying assumptions that reduce its real-world accuracy. The model uses a fixed gain factor (α) to establish a linear connection between water flow and humidity increase.

The actual relationship between water flow and humidity increase shows strong nonlinearity because it depends on various environmental factors. The model fails to consider the impact of water temperature on its operation. The amount of water vapor added to the air produces significantly different relative humidity increases when the air temperature is cooler compared to when it is warmer. The limitations of this model make GradientBoosting a more suitable machine learning approach for modeling real-time indoor humidity behavior, and offers a simplified yet meaningful explanation of the humidity dynamics observed during shower activity.

When the shower is running, the flow rate of water behaves like a pulse function—remaining constant during the shower and falling to zero afterward. In contrast, humidity typically increases in a more gradual manner. This can be modeled as the integral of the flow rate over time:

$$H(t) = H_0 + \int_0^t \alpha \cdot F(\tau) d\tau \quad (3.1)$$

where:

- $H(t)$ is the humidity at time t ,
- H_0 is the initial humidity level,
- $F(t)$ is the water flow rate in liters per minute (L/min),
- α is a gain constant translating water flow to humidity increase.

This relationship captures how continuous warm water flow leads to vapor generation and, consequently, a rise in humidity.

After the shower ends, relative humidity begins to dissipate through ventilation and natural air exchange. This phase can be described using a first-order differential equation:

$$\frac{dH(t)}{dt} = \alpha F(t) - \beta H(t) \quad (3.2)$$

Here, β is the decay constant representing the rate at which relative humidity is reduced due to fan activity or passive ventilation. This equation models the net change in relative humidity by balancing vapor addition (when flow is present) and relative humidity loss through ventilation.

By incorporating this rule-based model into our methodology, we establish a foundational reference that supports the interpretation of our predictive results. It also enables comparative analysis between machine learning forecasts and classical behavior models.

3.2 System Overview

The system uses multiple sensors and an actuator to automatically control a bathroom exhaust fan through ML-based relative humidity level predictions. The primary goal is to prevent mold growth while simultaneously lowering energy consumption and enhancing user comfort.

3.2.1 System Schematic

The schematic diagram in Figure 3.1 shows how the hardware components are interconnected to implement the smart relative humidity prediction and fan control system. It includes sensors (flow, temperature, humidity) that feed into the ESP32 microcontroller. Based on real-time data and TinyML predictions, the ESP32 controls the fan via a relay.

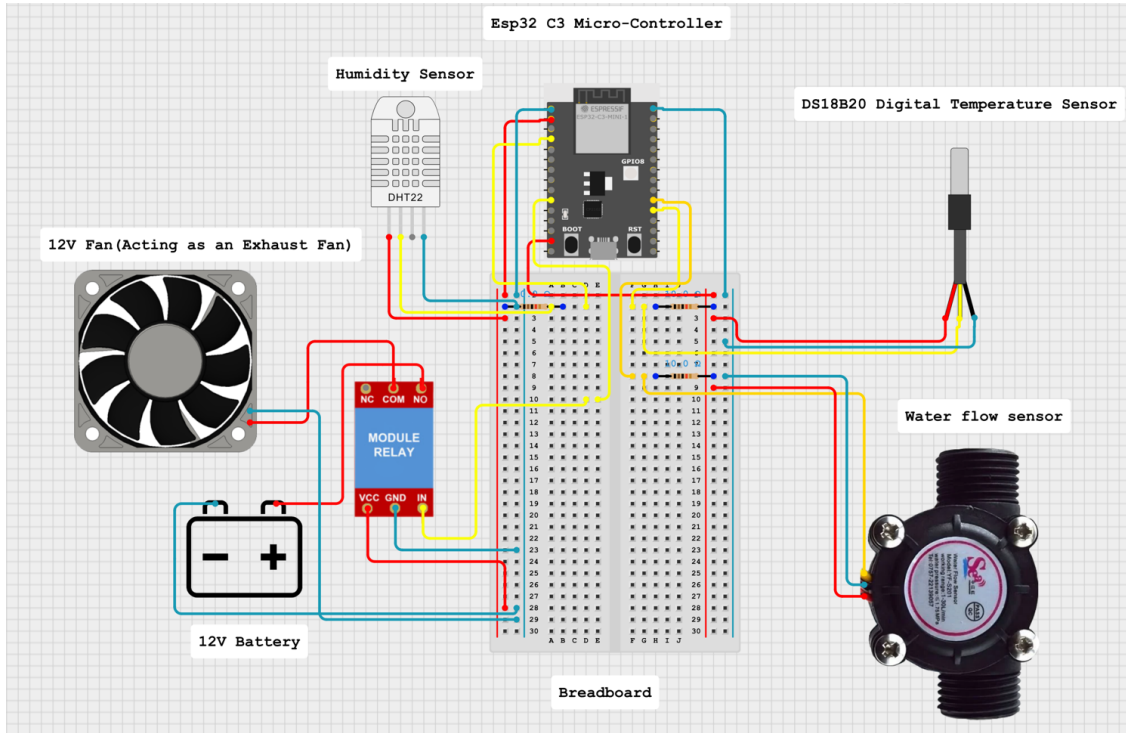


Figure 3.1: System schematic of smart relative humidity control setup

3.3 Working Principle

The system implements an intelligent relative humidity control mechanism using sensor fusion and TinyML-based prediction. Figure 3.2 illustrates the operational workflow.

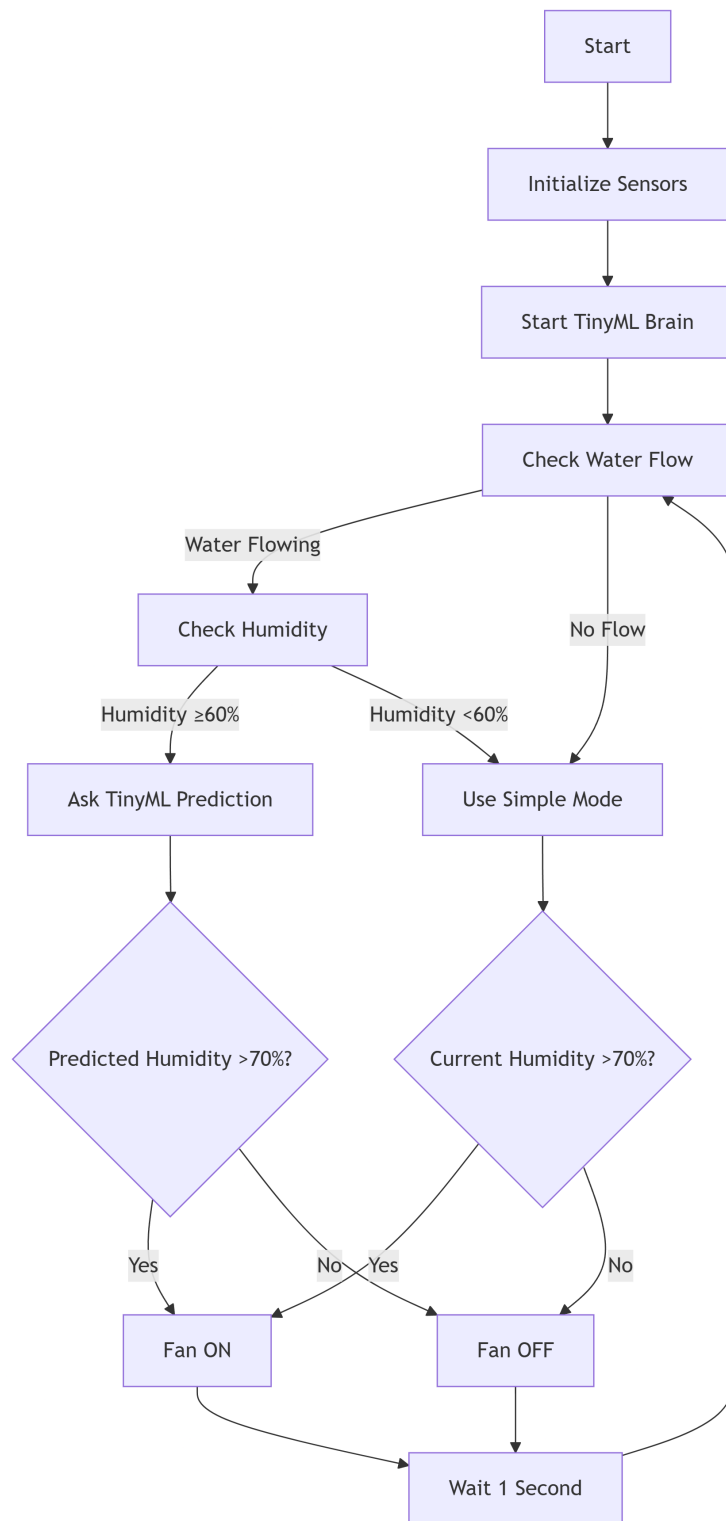


Figure 3.2: Flowchart of the smart relative humidity prediction and fan control system

System Initialization : Upon powering up, the ESP32 microcontroller initializes all connected components and prepares the system for real-time operation. The initialization process includes setting up the DHT22, DS18B20, and YF-S201 sensors,

which respectively monitor air humidity, measure water temperature, and detect water flow through interrupt pulse counting. These sensors form the sensory backbone of the system, providing the environmental parameters needed for intelligent control. Following sensor setup, the TensorFlow Lite Micro interpreter is initialized on the ESP32, loading the pre-trained TinyML model directly into memory. This model is designed to predict near-future relative humidity levels based on real-time readings of water flow and temperature. The initialization phase ensures that both the hardware and software environments are properly configured for seamless integration of sensor data acquisition and Machine Learning -driven decision-making, thereby laying the foundation for responsive and predictive ventilation control.

Main Loop Execution After initialization, the system enters a continuous loop where it performs real-time monitoring and decision-making. Initially, the ESP32 reads environmental data from the connected sensors. The DHT22 sensor captures the current air relative humidity as a percentage, the DS18B20 sensor records the temperature of flowing water in degrees Celsius, and the YF-S201 sensor calculates the flow rate in liters per minute using pulse counting. These readings form the input for the decision-making logic.

The system then evaluates the water flow condition. If water is detected flowing at a rate greater than 1 L/min, it checks whether the current relative humidity is equal to or exceeds 60%. If this condition is met, the system activates the TinyML-based prediction mode. If the relative humidity is below this threshold, or if no water flow is detected (i.e., flow rate is less than or equal to 1 L/min), the system defaults to a fallback mode where it relies on direct relative humidity checks instead of machine learning inference.

In the prediction mode, the flow rate and water temperature inputs are normalized to the ranges used during model training—specifically, flow rates between 10 and 20 L/min and temperatures between 20°C and 45°C are scaled to a [0, 1] interval. These scaled inputs are fed into the embedded TinyML model, which predicts the relative humidity level a few minutes into the future. If the predicted relative humidity exceeds 70%, the exhaust fan is turned on to prevent excessive moisture accumulation. Otherwise, the fan remains off to conserve energy.

In fallback mode, the system performs a direct evaluation of the current relative humidity level. If it exceeds 70%, the fan is activated; otherwise, it remains off. This ensures that the system retains basic functionality even when water flow ceases or the model inference fails. To maintain system stability and avoid overloading the microcontroller, the loop includes a one-second delay between iterations. This interval ensures consistent sampling and provides sufficient processing time for sensor reading, normalization, inference, and actuation logic.

3.3.1 Key Features & Logic

Table 3.1: Feature and its purposes

Feature	Purpose
TinyML Prediction	Uses AI to predict future humidity, reducing unnecessary fan usage.
Flow-Based Activation	ML only runs when water flows (saves power). RH
Fallback Mode	Ensures fan control even if ML/sensors fail. RH
Hysteresis (60%/70%)	Prevents rapid fan switching near threshold levels.
Sampling interval	10s = 10000ms

Operational Modes: The system operates through three distinct modes to ensure robust and adaptive relative humidity control. In predictive mode, the system becomes active when water flow exceeds 1 L/min and the ambient relative humidity is equal to or greater than 60%. Under these conditions, the TinyML model is engaged to forecast future relative humidity levels and make anticipatory fan control decisions. When no water flow is detected, the system transitions to fallback mode, relying on direct relative humidity measurements to determine whether fan activation is necessary.

Additionally, the system includes a safety mode that automatically deactivates control functions if sensor faults are detected, such as the occurrence of non-numeric (NaN) values, thereby ensuring reliability and operational integrity. The system reduces unnecessary computation by first verifying that the measured relative humidity exceeds 60% before activating the prediction model. This condition helps to save processing power and prevent false alarms during low relative humidity conditions.

Example Scenario: In a typical use case, the system demonstrates its predictive and fallback control mechanisms. When water is flowing at a rate of 2 L/min and the relative humidity reaches 65%, the embedded TinyML model predicts a future humidity level of 72%. Based on this forecast, the fan is automatically activated to prevent excessive moisture buildup. Once the water flow ceases, dropping to 0 L/min, and the relative humidity is measured at 68%, the system transitions into fallback mode. Since the relative humidity remains below the critical 70% threshold, the fan remains off.

However, if the relative humidity continues to rise and reaches 75% despite no water flow, the fallback mechanism activates the fan to maintain optimal air quality. This scenario highlights the system’s ability to anticipate relative humidity surges and maintain control even in the absence of active water usage, ensuring robust and energy-efficient ventilation.

3.4 Component Selection and Justification

- **Temperature Sensor (DS18B20):**

The DS18B20 Temperature Sensor 3.3 was selected because of its digital precision and reliability in humid conditions. [12]



Figure 3.3: DS18B20 temperature sensor

- **Water Flow Sensor (YF-S201):**

The YF-S201 Water Flow Sensor 3.4 was used to measure the current water consumption which is essential for determining bathing incidents. [11]



Figure 3.4: YF-S201 water flow sensor

- **Humidity Sensor (DHT22):**

The DHT22 Humidity Sensor 3.5 provides actual data for training and validating TinyML models. [13]

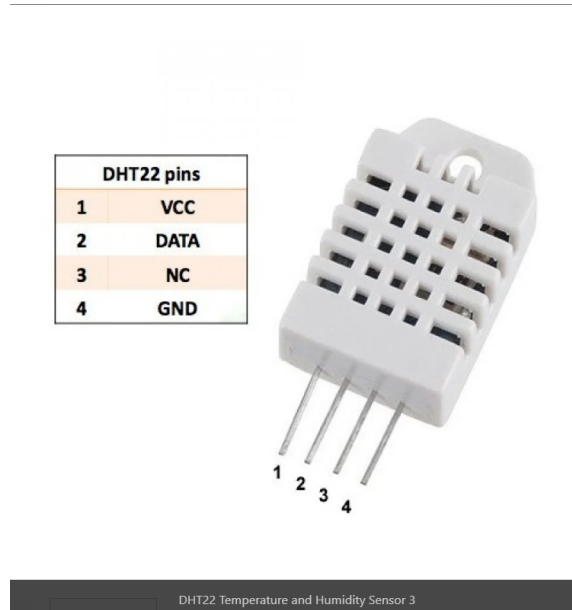


Figure 3.5: DHT22 humidity sensor

- **ESP32 Microcontroller:**

The ESP32 Microcontroller3.6 provides enough processing power and storage to execute TinyML models and it supports Wi-Fi connectivity for data logging. [23]

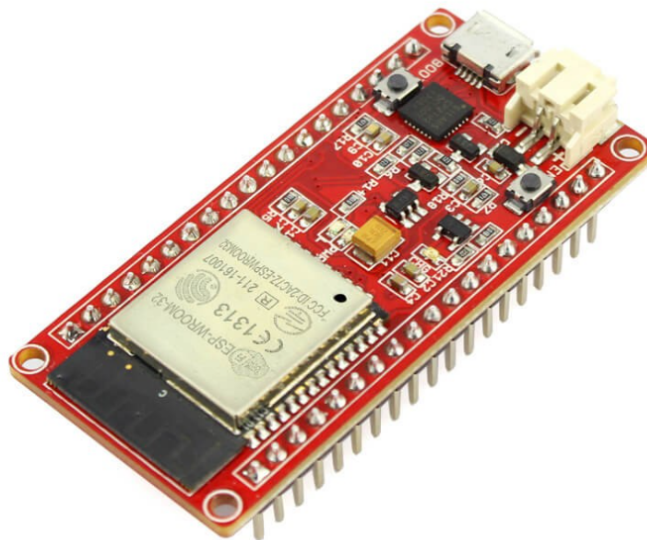


Figure 3.6: ESP32 microcontroller

- **Exhaust fan:**

The system can control air flow through the relay-controlled exhaust fan based on predicted humidity levels. This combination enables real-time control of

multiple parameters without any external dependency.3.7



Figure 3.7: Exhaust fan

- **Relay Module for Fan Control:**

The relay module 3.8 is used to switch the exhaust fan on and off based on predictions from the embedded TinyML model. It acts as an electrical bridge between the low-power ESP32 microcontroller and the high-voltage fan, enabling automated control without manual intervention

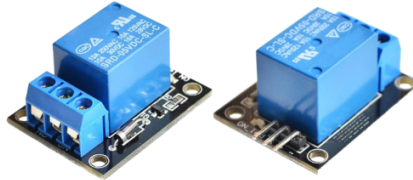


Figure 3.8: Relay module used for exhaust fan control

3.4.1 Sensor Configuration and Setup

An ESP32 microcontroller was paired with three sensors. These sensors were selected due to their support on embedded platforms and robustness in humid environments. We programmed the ESP32 to sample all three sensors every 10 seconds, coordinated with the real-time inference loop of the running system. Sensor measurements were captured via serial communication to a networked PC.

3.4.2 Data Collection Conditions

We collected data for three consecutive days under actual usage scenarios to ensure the model was trained on realistic environmental conditions. The data included observations before, during, and after showering to capture the full relative humidity

progression within the bathroom environment. Additionally, we monitored idle periods characterized by zero water flow, during which no one was using the shower, to establish baseline relative humidity and temperature values. This comprehensive data collection approach enabled us to observe a wide range of operating states, including abrupt relative humidity spikes and transitional phases, thereby enhancing the robustness of the predictive model.

3.5 Justification of Chosen Method

The evaluation of traditional rule-based control strategies, including fixed relative humidity setpoints and cloud-based TinyML models, revealed significant limitations in terms of adaptability, responsiveness, and user privacy. Threshold-based control systems are inherently rigid and unable to dynamically adjust to fluctuations in environmental conditions or individual usage patterns. Similarly, cloud-based inference mechanisms introduce considerable latency and pose privacy risks, as they require constant internet connectivity and data transmission to external servers.

In contrast, TinyML provides a compelling alternative for embedded applications by enabling local, real-time inference on low-power microcontrollers. This approach offers reduced latency, improved energy efficiency, and full autonomy from external cloud servers, thereby enhancing data privacy and system responsiveness. Furthermore, the ability to embed learned patterns within the device increases system adaptability and performance over time. The combination of these advantages makes TinyML an ideal solution for resource-constrained, privacy-sensitive use cases such as intelligent bathroom automation.

3.6 Validation and Testing

The system was evaluated with a complete hardware setup to ascertain its predictability and responsiveness in actual bathroom use conditions.

Evaluation Framework

The TinyML model was embedded on an ESP32 microcontroller to facilitate local predictions based on data acquired from sensors monitoring water flow, temperature, and humidity. Real-time testing was conducted with varying shower durations and room temperatures to simulate realistic usage conditions. The predicted relative humidity readings were then compared with the actual values measured by the DHT22 sensor to assess the model’s accuracy and responsiveness.

Model Evaluation Metrics

To evaluate the performance of machine learning models for relative humidity prediction, we used the following standard regression metrics:

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|. \quad (3.3)$$

Measures the average magnitude of errors in predictions, regardless of direction.

- **Mean Squared Error (MSE):**

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (3.4)$$

Penalizes larger errors more than MAE due to squaring.

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\text{MSE}}. \quad (3.5)$$

Expresses error in the same units as the target variable (Relative Humidity).

- **Coefficient of Determination (R^2):**

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (3.6)$$

Indicates the proportion of variance explained by the model. $R^2 = 1$ indicates perfect prediction.

Fan Control Validation

- The fan was turned on in expectation of increased relative humidity during rain, thereby confirming that the predictive logic functioned as intended.
- The fan control logic achieved perfect classification accuracy (Precision, Recall, and F1 Score = 1.00), verifying its reliability under test scenarios.

Overview

The testing phase validated that the embedded system was good at predicting relative humidity rises and acted by regulating the fan in real time. The model functioned well within the ESP32 hardware limitations, demonstrating its applicability in embedded predictive control with no internet reliance.

3.7 Validity and Reliability Considerations

We ensured the validity and reliability of our system through a structured approach involving sensor calibration, rigorous testing, and controlled experimentation. The DHT22 and DS18B20 sensors were pre-calibrated and cross-verified against commercial-grade hygrometers and thermometers to confirm measurement accuracy. To increase data reliability, we conducted multiple test runs under identical conditions, which helped to reduce overfitting and minimize random noise in sensor readings. Regarding internal validity, the cause-and-effect relationship between water consumption and rising relative humidity levels was established within a controlled experimental setup to isolate key influencing variables. However, the generalizability of our findings is subject to external validity constraints. Although the system demonstrated reliable performance in a standard residential bathroom, factors such as ventilation duct dimensions, room volume, and insulation characteristics may affect its applicability in different bathroom environments. These considerations were essential in assessing the robustness and transferability of our relative humidity prediction and fan control framework.

3.8 Limitations and Self-Critique

The system we developed, while functional, has several limitations that should be acknowledged. The TinyML model was trained on a relatively small dataset collected from a single bathroom environment, which may impact its performance when deployed in different bathroom layouts or under varying environmental conditions. The current implementation relies on only two predictors—flow rate and water temperature—while actual relative humidity accumulation is influenced by several other factors such as external weather conditions and the level of insulation in the room. Furthermore, battery usage and energy efficiency were not investigated, which limits the system’s portability and long-term deployment potential. For future iterations, it will be important to incorporate auto-calibration mechanisms, expand the range of predictive features, and include occupancy detection capabilities to provide better contextual awareness and enhance system adaptability.

3.9 Conclusion of Methodology

The methodology is designed to answer the central research questions by integrating AI-based prediction with embedded systems. Through careful selection of components, sensor calibration, TinyML integration, and systematic validation, the system demonstrates a novel and practical approach to autonomous relative humidity control in bathrooms.

4.1 Training Data for the Model

To train the TinyML model embedded within the device for relative humidity forecasting, we collected sensor data in a real-world residential bathroom environment. The objective was to record how relative humidity levels vary in response to shower usage as a function of water flow and temperature changes. To validate the theoretical system design, a functional prototype was developed using the ESP32 microcontroller, DHT22 relative humidity sensor, DS18B20 temperature sensor, and YF-S201 water flow sensor. These components were assembled on a breadboard and connected with jumper wires, with the exhaust fan controlled via a relay module. The physical setup of the prototype is shown in Figure 4.1, while Figure 4.2 depicts live testing conducted in a residential bathroom setting to evaluate the system’s performance under real-world environmental conditions.

The recorded data helped us capture relative humidity behavior across different phases: pre-shower, active showering, and post-shower, along with idle periods where no water flow occurred. This enabled the model to observe natural transitions and identify patterns leading to relative humidity spikes. Figure 4.3 illustrates how relative humidity levels change over time and how the fan activation correlates with shower activity. The top plot in the figure presents relative humidity (%) as a smooth green line, helping track its rise during showering and decline during active ventilation. The bottom plot uses colored bands—blue for shower activity and purple for fan operation—to mark event timelines. These plots share a common time axis, allowing a clear view of how sensor-detected events align with environmental changes.



Figure 4.1: Top view of the hardware prototype: ESP32, sensors, and relay module

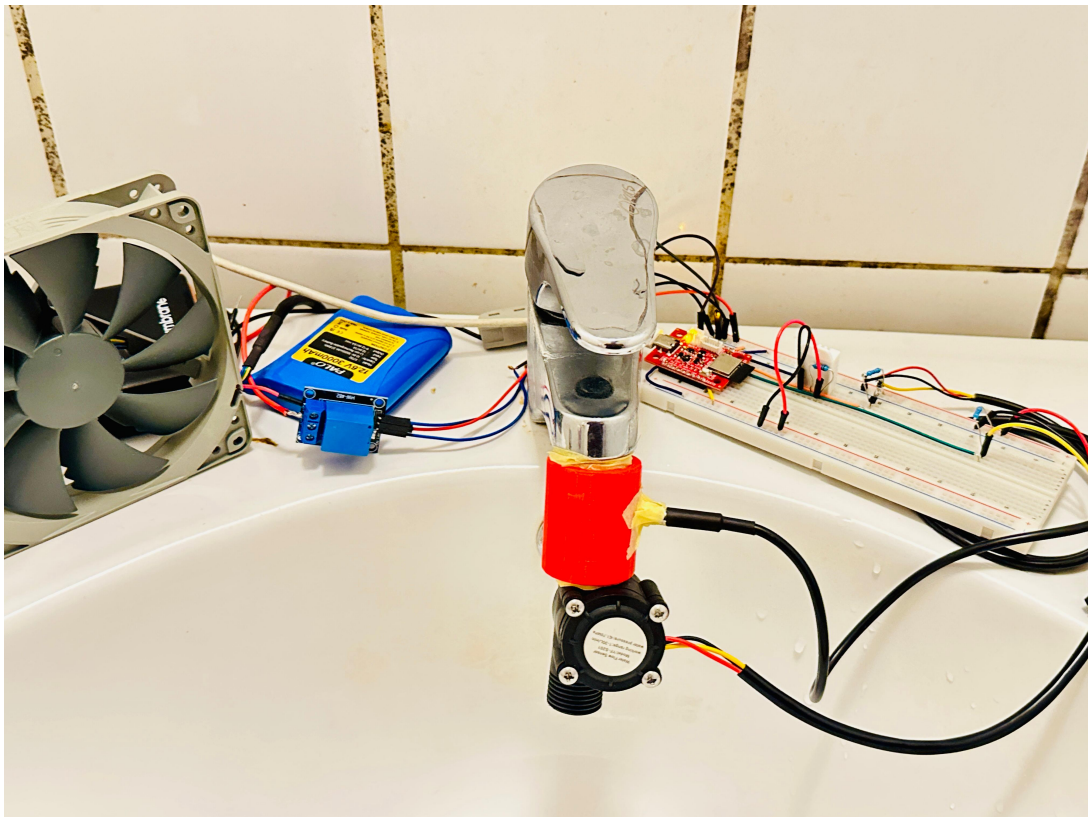


Figure 4.2: Live testing of the prototype in a bathroom setup

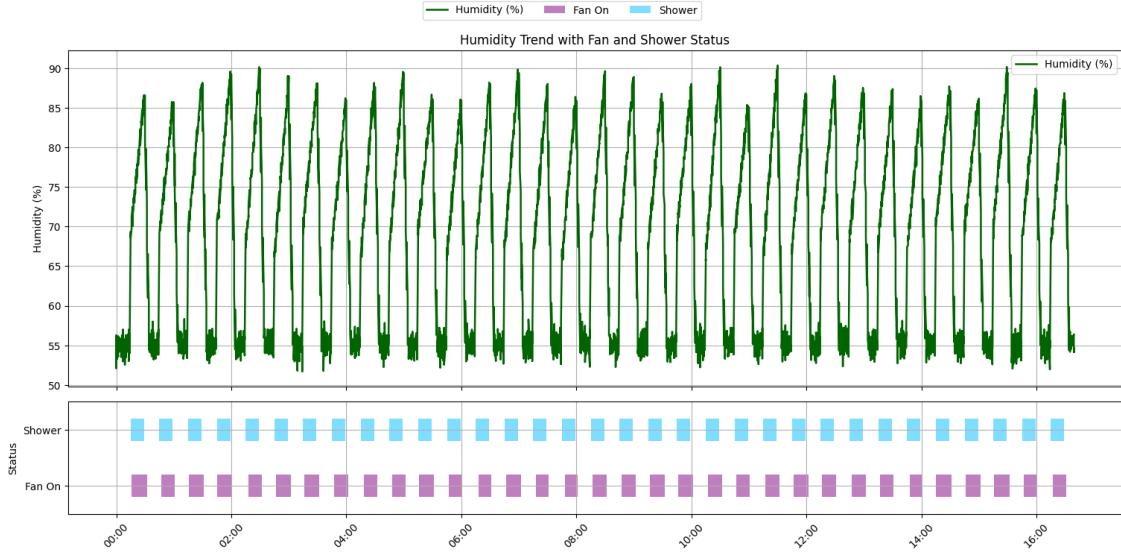


Figure 4.3: relative humidity vs. Time showing shower and fan activation phases.

4.1.1 Dataset Characteristics

In total, we collected 6000 usable samples. Each sample included:

- Water flow rate,
- Water temperature,
- Ambient humidity.

Even in its limited duration, this data was sufficient to cover short-term relative humidity behavior relevant to bathroom use. It was the suitable size to train a light model to execute in real-time on the ESP32.

4.1.2 Preprocessing and Labeling

Before training the machine learning models, we normalized all input features to a common numerical scale to ensure consistency in performance across different algorithms. The flow rate was scaled from its observed range of 10–20 L/min to the interval $[0, 1]$, based on empirical data recorded during typical shower usage. Similarly, water temperature values ranging from 20°C to 45°C were normalized to the same scale. Ambient relative humidity was used as a reference input without scaling, as its native range of 40–90% was already compatible with the model’s internal handling and did not require transformation.

These preprocessing steps were consistently applied across all four machine learning models—Linear Regression, Random Forest, Gradient Boosting, and Support Vector Regression—to maintain numerical stability during training. The target output for all models was the forecasted relative humidity value five minutes into the future. We achieved this by shifting the relative humidity readings in time, aligning each input sample with the corresponding future relative humidity value. This preprocessing approach enabled anticipatory rather than reactive predictions.

All data used in training was collected locally within our bathroom environment, and no external datasets or cloud-based augmentations were included. Processing was done entirely offline to protect user privacy and ensure that the embedded system remained self-contained. During deployment on the ESP32, the same normalization procedures were implemented to ensure consistency between the training and inference environments.

4.1.3 Summary

Our data collection methodology was directed towards capturing meaningful environmental changes during normal bathroom usage. By combining real-time water flow, temperature, and relative humidity readings, we created a focused and representative dataset. The dataset allowed us to create a small and deployable TinyML model for smart, embedded humidity prediction.

4.1.4 Feature Correlation Analysis

To understand the interrelationships among the collected sensor variables and their influence on exhaust fan activation, we conducted a feature correlation analysis using Pearson correlation coefficients, visualized through a heatmap in Figure 4.4. This analysis served as an important step in evaluating the suitability of sensor data for predictive modeling, particularly for forecasting relative humidity levels and triggering timely ventilation [24, 25].

The dataset used in this analysis consisted of approximately 6,000 time-synchronized samples collected over a three-day period from a real residential bathroom equipped with an ESP32-based sensor system. Variables included ambient humidity, water flow rate, water temperature, and the binary status of the exhaust fan. All data were recorded in real time during various user interactions, including shower activity and idle bathroom states. Prior to correlation computation, the dataset was normalized to ensure scale invariance. The Pearson correlation matrix was calculated using the `.corr()` function from the `pandas` library, and the heatmap was generated with `seaborn.heatmap()` to facilitate clear visualization.

The results showed a strong positive correlation (0.90) between relative humidity and fan status, affirming that elevated relative humidity levels are a primary factor in triggering exhaust fan activation. Flow rate and water temperature also demonstrated moderate positive correlations with fan status, both approximately in the 0.68 to 0.69 range. These findings suggest that while relative humidity is the dominant predictor, water flow and temperature also carry significant predictive value. Interestingly, we also observed a very high correlation of 0.97 between water flow rate and water temperature, likely due to their common dependence on shower usage behavior. This multicollinearity is particularly relevant for linear modeling techniques but is less problematic in ensemble methods like Gradient Boosting, which we later selected due to their robustness to such feature dependencies.

Overall, this correlation analysis informed our model design by validating the inclusion of all three sensor inputs in our supervised learning pipeline. It reinforced the relevance of using sensor fusion for accurate and anticipatory relative humidity prediction and provided early justification for employing machine learning techniques

capable of managing complex feature interactions.

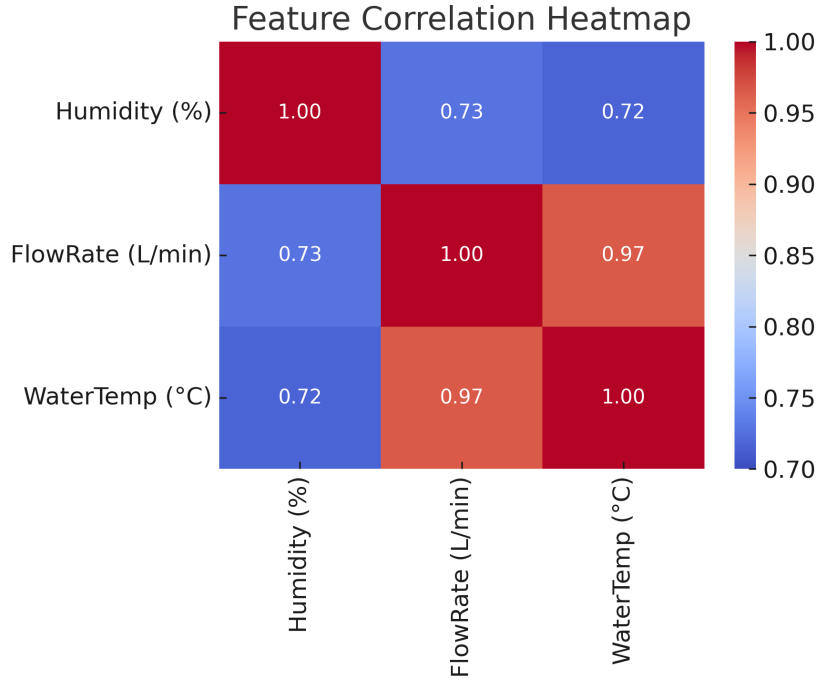


Figure 4.4: Heatmap showing Pearson correlation coefficients among humidity, flow rate, and water temperature, .

The model input does not use FanStatus but its correlation with relative humidity (0.9) was analyzed for system behavior verification. The high correlation is expected because the fan is programmed to turn ON when relative humidity exceeds a certain threshold. Therefore, the FanStatus signal acts more as a consequence of high relative humidity rather than a cause. This supports the system logic and shows that fan activation reliably responds to elevated relative humidity levels, as designed.

4.1.5 3D-Printed Prototype Design and Bathroom Integration

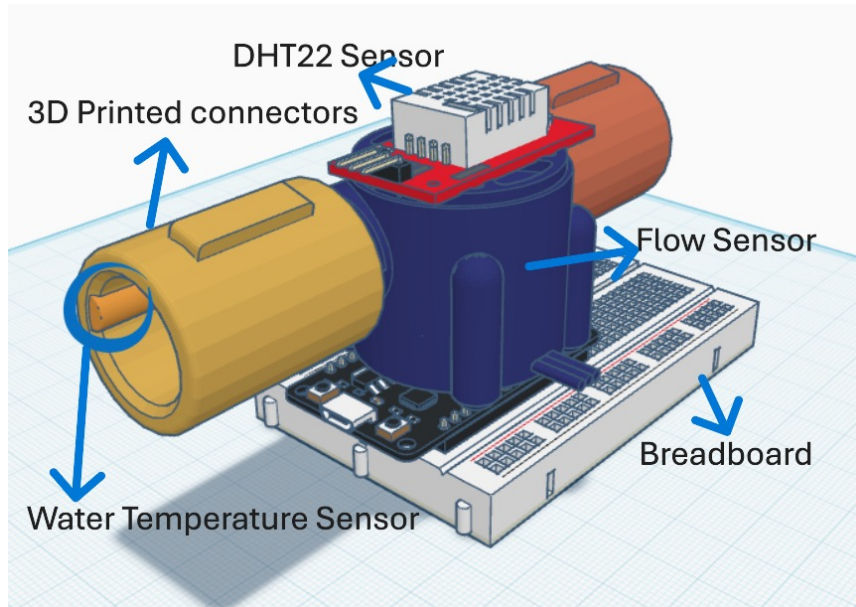
To bring our design from concept to reality, we built a 3D-printed prototype tailored for easy, non-intrusive integration into a typical shower setup. It houses all key sensors—flow rate, water temperature, and humidity—in a compact cartridge that fits seamlessly into existing plumbing without any permanent modifications.

Figure 4.5a shows a close-up of the sensor assembly. Custom 3D-printed connectors in PLA/ABS link the sensors directly to standard shower hoses. A DS18B20 probe monitors water temperature, a YF-S201 sensor measures water flow, and a DHT22 records ambient humidity. All sensor data is processed by an ESP32 microcontroller mounted on a half-sized breadboard.

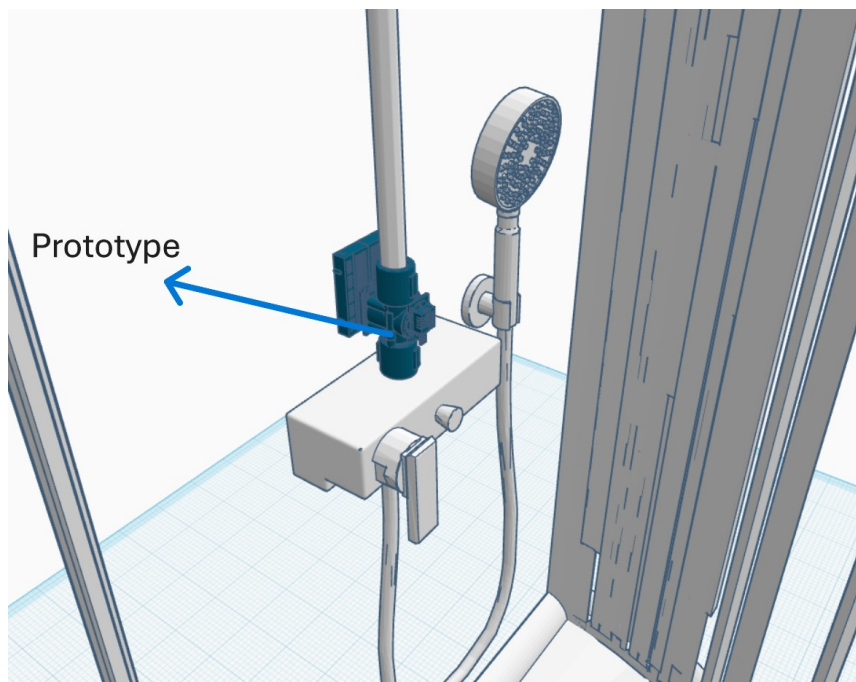
In Figure 4.5b, the assembled unit is mounted on the shower mixer using 3D-printed sleeves for stable, leak-free alignment. The compact design keeps all components accessible for firmware updates without obstructing shower use.

The updated system context is shown in Figure 4.6, now including the exhaust fan mounted on the stall wall. The waterproof sensor module is placed just before the diverter valve to monitor all water flow. The fan automatically activates based on predicted relative humidity levels, effectively ventilating moist air outside the bathroom to prevent mold buildup.

This setup confirms our design goals: it is easy to install, reversible, and mechanically stable. The use of a breadboard also accelerated testing cycles prior to migrating to a custom PCB, demonstrating real-world readiness of our TinyML-powered monitoring system.



(a) Close-up of the sensor assembly showing water temperature, flow sensor, DHT22, and breadboard-mounted ESP32.



(b) Prototype assembly mounted on the mixer block with 3D-printed alignment sleeves.

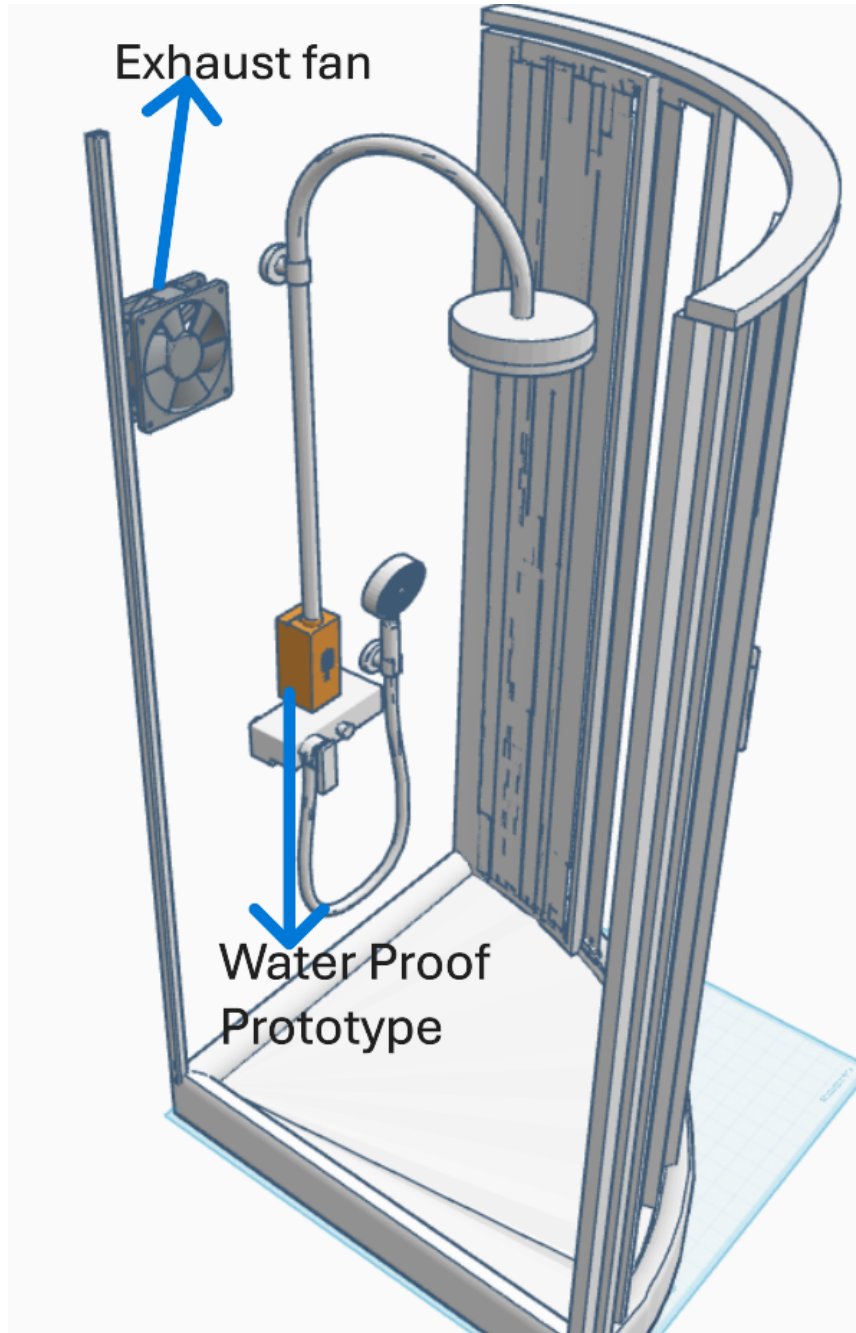


Figure 4.6: Full system setup inside a shower stall, showing the waterproof prototype and exhaust fan for humidity ventilation.

4.2 Model Performance

We evaluated the performance of four different machine learning models—Linear Regression, Random Forest, Gradient Boosting, and Support Vector Regression—for forecasting relative humidity levels based on input features such as water flow rate, water temperature, and current humidity. Each model was assessed using standard regression metrics including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the Coefficient of Determination

(R^2), defined in Equations 3.3, 3.4, 3.5, and 3.6 respectively. These metrics helped quantify both average and variance-based deviations between predicted and actual relative humidity values.

Among the tested models, GB outperformed others with the highest accuracy ($R^2 = 0.86$, $MAE = 3.31$). Its capability to execute directly on the ESP32 ensured fast, private, and proactive control of humidity, enabling preemptive fan activation. Forecasting focused on short-term intervals (2–5 minutes) due to observed rapid relative humidity rise during initial shower periods. A 5-minute prediction horizon was implemented by shifting training labels 300 seconds ahead, aligning inputs at time t with relative humidity values at $t + 300$. This supervised setup allowed the model to learn near-future relative humidity conditions from current sensor readings. Longer forecast intervals reduced accuracy due to increased uncertainty, justifying the chosen short-term strategy.

As summarized in Table 4.1, Gradient Boosting outperformed all other models, achieving the lowest MAE of 3.31 and RMSE of 4.35. This indicates better prediction accuracy and lower error margins compared to the other tested models. Furthermore, its R^2 score of 0.86 suggests that approximately 86% of the variance in relative humidity levels was successfully captured by the model, highlighting its effectiveness in modeling the underlying relationship between environmental parameters and short-term relative humidity trends.

Table 4.1: Model performance comparison

Model	MAE	MSE	RMSE	R^2
Linear Regression	3.98	23.36	4.83	0.82
Random Forest	3.53	22.34	4.73	0.83
Gradient Boosting	3.31	18.97	4.35	0.86
SVR	3.14	19.14	4.37	0.85

4.3 Gradient Boosting Time Series Prediction

To evaluate the effectiveness of our predictive model, we focused on Gradient Boosting for short-term relative humidity forecasting. The plot shown in Figure 4.7 demonstrates the model’s performance in predicting relative humidity levels five minutes in advance. The predicted values, represented by the orange dashed line, closely follow the actual relative humidity measurements shown as a solid green line. This close alignment illustrates the high predictive accuracy achieved by the model.

For model training, we calculated that five minutes corresponds to 300 seconds, and since sensor readings were collected at 10-second intervals, this resulted in a prediction offset of 30 samples. Therefore, input features from time t were paired with target relative humidity values from time $t + 30$. This formulation allowed the model to learn the dynamics of relative humidity change in advance of actual shifts.

As demonstrated in Figure 4.7, the model accurately captures the rapid rise and fall in relative humidity associated with typical bathroom usage. The minimal lag between prediction and real values highlights its responsiveness to dynamic environmental changes. Furthermore, the model’s anticipatory capability supports energy

efficiency by enabling timely activation of the exhaust fan, reducing unnecessary usage. These implications are further discussed in the system’s broader context within the discussion chapter.

In addition to forecasting performance, the relationship between actual and predicted relative humidity values is visualized in Figure 4.8. The data points cluster closely along the 45° reference line, indicating strong alignment between observed and forecasted values. A high correlation coefficient of 0.926 further validates the model’s accuracy.

The environmental behavior of key features such as flow rate and water temperature is also noteworthy. As shown in Figure 4.9, there is a moderate positive correlation between water flow and temperature, indicating that higher flow rates typically coincide with elevated temperatures during shower use. Furthermore, flow rate exhibited a bimodal distribution with peaks around 0 L/min and 12 L/min, corresponding to idle and active shower states. This distribution is presented in the histogram of Figure 4.10.

Figure 4.3, introduced earlier, reinforces that relative humidity levels sharply increase during active showering periods, shown by the purple region, and decrease during fan operation, marked by the red region. This confirms the hardware’s real-time decision-making and the utility of the forecasting model in enabling energy-efficient ventilation control.

We observed that **Figure 4.9** (During shower: high flow rate and elevated water temperature) displays data specifically from the active shower period. Contrary to earlier assumptions, this figure does *not* reflect ambient air temperature during idle periods when water flow is absent. Therefore, the previously claimed temperature range at 24°C, intended to represent idle bathroom conditions, is not supported by this figure and should not be inferred from it.

Instead, the plot shows a loosely scattered distribution between flow rate and water temperature during shower usage. While the overall dataset analysis yielded a strong Pearson correlation of 0.97 between these two variables, this subset illustrates how that relationship weakens in real-time during showers. The apparent discrepancy highlights how correlations observed across the entire dataset may not hold within narrower time windows or specific behavioral contexts.

This scatter may result from varying user preferences, adjustment phases in water heating, or delayed stabilization of temperature sensors relative to flow initiation. Such variability implies that although flow rate and temperature appear statistically correlated overall, they may behave more independently in real-time use. This insight is critical for developing robust prediction models—these features should be treated as individually informative rather than tightly coupled.

In future analyses, it would be beneficial to visualize these variables as time-aligned series over entire shower sessions to better observe their dynamic interactions. For now, this scatter plot serves to caution against over-reliance on summary statistics without examining contextual data behavior.

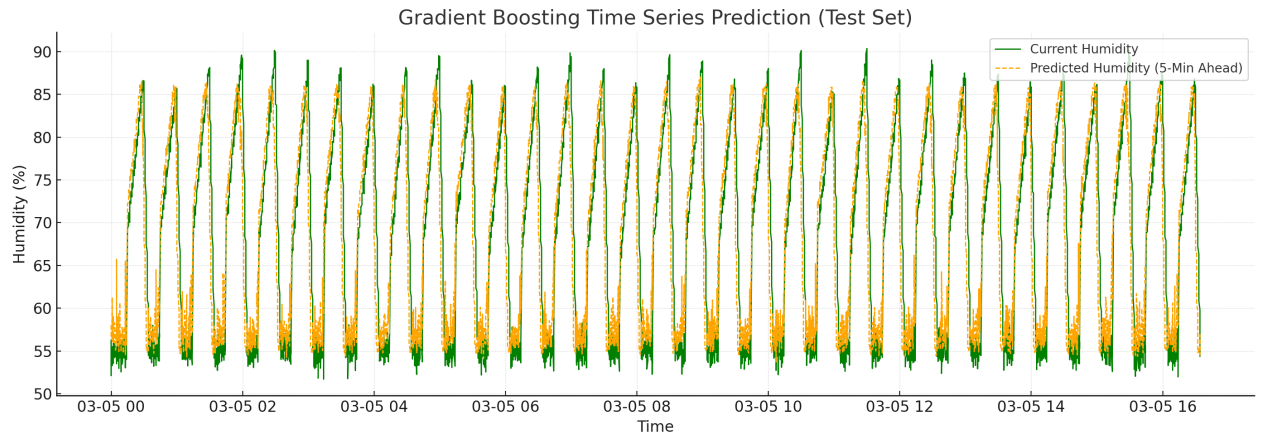


Figure 4.7: Gradient Boosting Time Series Prediction showing actual and forecasted relative humidity (Test Set)

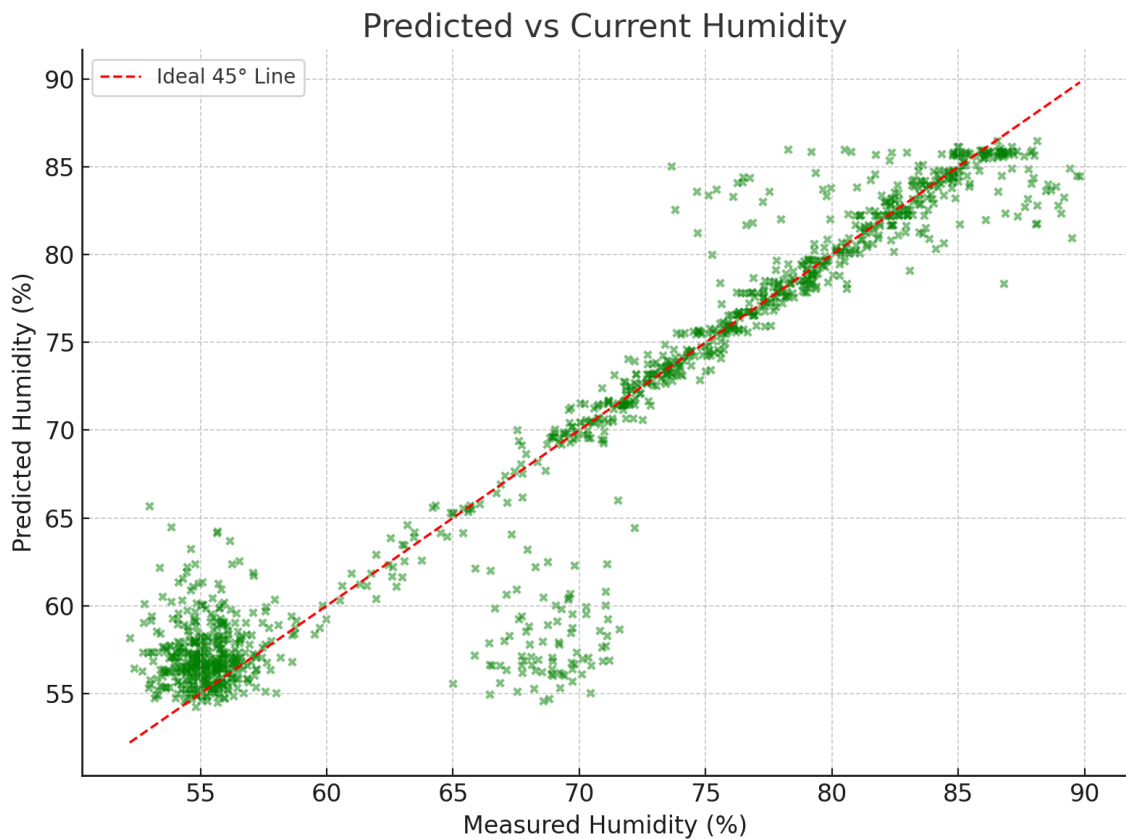


Figure 4.8: Actual vs. Predicted relative humidity with 45° reference line

The predicted relative humidity values appear in Figure 4.8 alongside measured relative humidity values with the 45° reference line indicating perfect prediction. The model shows excellent overall accuracy especially in the 70% to 85% relative humidity range where most points closely follow the reference line. The model demonstrates high reliability in predicting peak relative humidity which serves as the critical zone during showery events. The model shows noticeable clusters in both the lower and

mid-humidity ranges. The model shows higher predicted values than actual measurements below 60% relative humidity because it learns to detect rising relative humidity at the start of shower cycles thus becoming more cautious. The transitional phase of relative humidity increase between 65% and 70% produces a second cluster with increased data spread because input conditions become less clear during this period. The model demonstrates consistent behavior with proactive control objectives even though it shows slightly reduced precision in these specific areas. The model focuses on early detection which supports its objective of preventive ventilation and energy efficiency although it sometimes overestimates during periods of low risk.

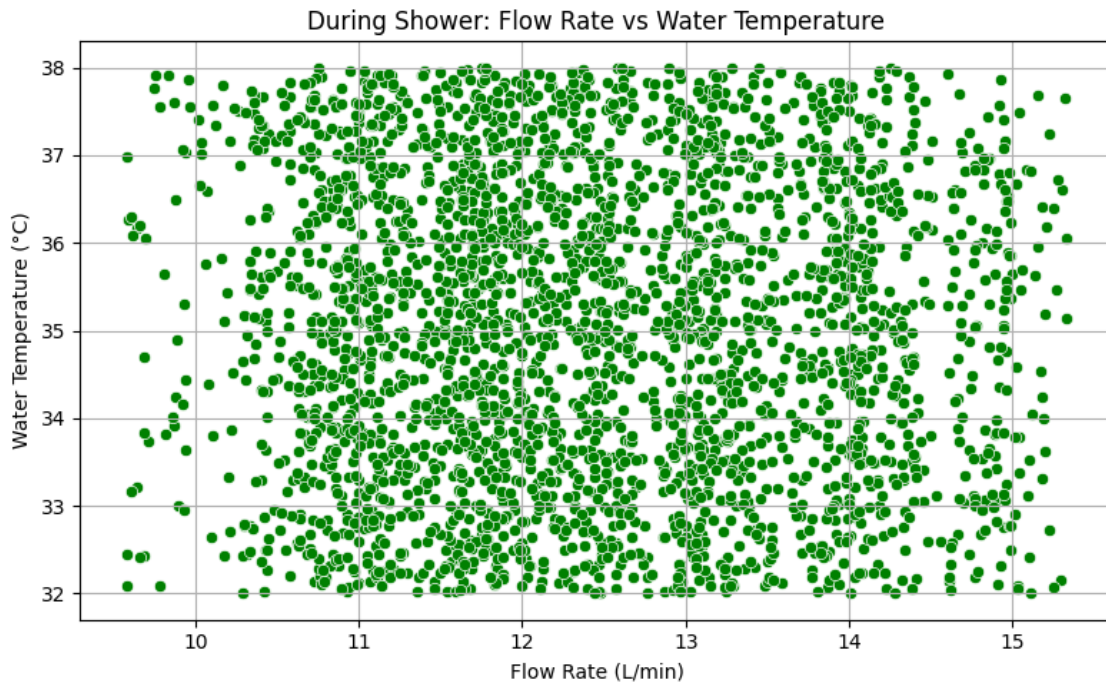


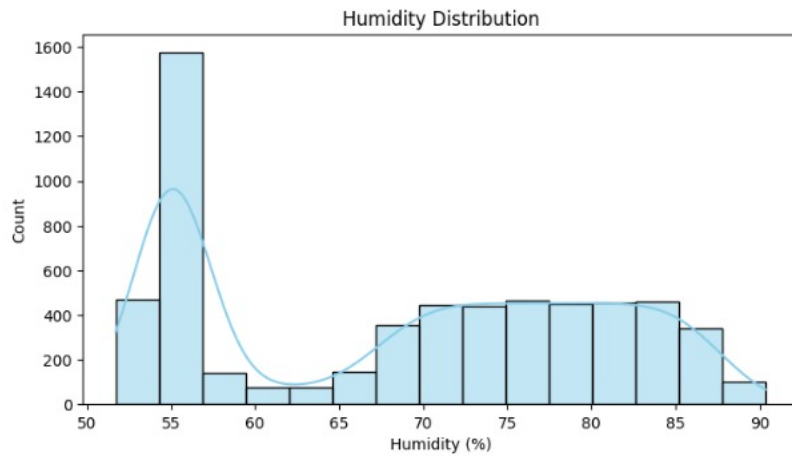
Figure 4.9: Scatterplot showing flow rate vs. water temperature during showers. Scatter suggests phase-wise or user-level variability despite high overall correlation.

The scatterplot in Figure 4.9 shows the relationship between low rate and water temperature during shower activity. The correlation matrix showed a very strong linear relationship between these two variables (correlation = 0.97), but the visual representation here appears more scattered and less clearly linear. This inconsistency suggests that while the overall dataset supports a high correlation, the relationship may not be consistently strong during all shower phases.

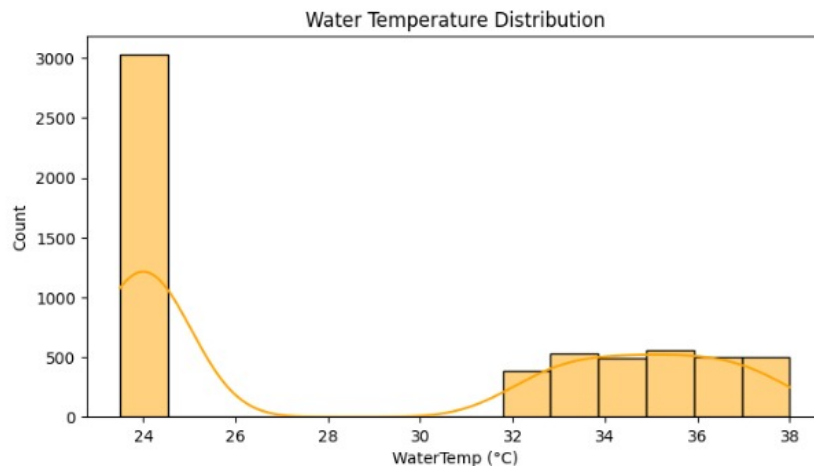
For example, the initial start of a shower may involve low flow and rising temperature, while the middle phase may stabilize both, and the end may show flow tapering off with residual heat. User specific behavioral also plays a role and different users may prefer different temperature and pressure combinations. These variations could contribute to scatter in the data despite a strong overall statistical correlation.

This plot highlights the importance of considering contextual phases and user patterns in interpreting sensor relationships and reminds us that visualizations can reveal dynamics that single metrics like Pearson correlation may obscure. It also suggests that Flow Rate and Water Temperature, while correlated overall, should

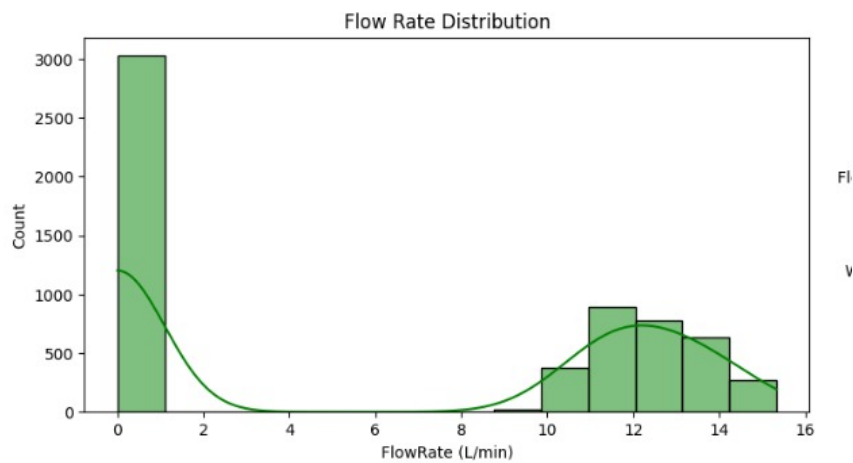
still be treated as distinct inputs in modeling and not assumed to move together in all operational contexts.



(a) Humidity



(b) Flow rate (L/min)



(c) Water temperature (°C)

Figure 4.10: Histograms of humidity, flow rate, and water temperature.

4.3.1 Sample-Based Validation of Fan Control Logic

The behavior of both reactive and predictive fan control logic was visually validated by analyzing a sample sequence from the dataset.

Timestamp	FlowRate (L/min)	WaterTemp (°C)	Humidity (%)	FanStatus(Measured)	Predicted Humidity (%)	FanStatus(Predicted)
03-05-2025 00:12	0	23.58	54.99	0	63.66062612	0
03-05-2025 00:12	0	24.35	56.34	0	62.78888338	0
03-05-2025 00:12	0	23.91	54.18	0	61.72856046	0
03-05-2025 00:13	0	24.38	54.09	0	61.87439581	0
03-05-2025 00:13	0	24.49	56	0	63.19200299	0
03-05-2025 00:13	0	24.36	54.16	0	61.13668013	0
03-05-2025 00:13	0	24.46	53.03	0	62.43416917	0
03-05-2025 00:13	0	24.45	54.32	0	61.02940236	0
03-05-2025 00:13	0	24.37	56.98	0	61.68466888	0
03-05-2025 00:14	0	24.34	54.35	0	60.79634447	0
03-05-2025 00:14	0	23.94	54.02	0	62.40316981	0
03-05-2025 00:14	0	24.43	55.01	0	62.71193573	0
03-05-2025 00:14	0	24.45	56.24	0	63.0159426	0
03-05-2025 00:14	0	23.89	56.34	0	63.30054125	0
03-05-2025 00:14	0	24.23	55.82	0	63.08104831	0
03-05-2025 00:15	14.97	35.89	68.88	0	75.66097994	1
03-05-2025 00:15	14.02	37.68	68.75	0	75.82528647	1
03-05-2025 00:15	13.88	32.48	68.58	0	75.74764936	1
03-05-2025 00:15	14.13	33.23	68.94	0	75.84201412	1
03-05-2025 00:15	14.16	33.49	68.5	0	75.70096516	1
03-05-2025 00:15	14.91	34.67	69.35	0	76.26685973	1
03-05-2025 00:16	14.2	35.23	69.71	0	76.40925554	1
03-05-2025 00:16	13.87	36.98	69.94	0	76.73012276	1
03-05-2025 00:16	13.89	35.27	70.42	1	77.29772891	1
03-05-2025 00:16	13.89	36.06	70.76	1	77.74596442	1
03-05-2025 00:16	14.04	32.87	70.91	1	77.6438568	1
03-05-2025 00:16	14.99	35.37	71.15	1	77.78805209	1
03-05-2025 00:17	14.18	33	70.04	1	77.24010923	1
03-05-2025 00:17	14.25	35.2	71.3	1	78.05066138	1
03-05-2025 00:17	14.93	32.31	71.4	1	77.90826558	1
03-05-2025 00:17	14.71	36.82	71.53	1	78.04210744	1
03-05-2025 00:17	13.97	35.8	71.99	1	79.05435405	1
03-05-2025 00:17	14.68	35.09	72.52	1	79.24887086	1
03-05-2025 00:18	14	33.59	72.22	1	78.95224643	1
03-05-2025 00:18	13.99	37.65	72.81	1	79.84227669	1
03-05-2025 00:18	15.03	32.09	73.09	1	79.49247193	1

Figure 4.11: Fan activation behavior based on measured and predicted relative humidity values during a sample shower cycle

The predictive control logic effectiveness was validated through an analysis of a sample from the system's time-series dataset. The following table shows readings taken during a typical shower session which includes timestamped values of flow rate, water temperature, measured humidity, predicted relative humidity and fan activation status under both reactive and predictive modes.

The initial readings showed both measured and predicted relative humidity values stayed between 54–56% when the bathroom was unoccupied (FlowRate=0). The fan operated in OFF mode in both reactive and predictive control modes.

The system detected a major increase in water flow (14–15L/min) and temperature (35°C) at 00:14 which indicated the beginning of a shower. The reactive fan mode activated at 00:16 when measured relative humidity exceeded 70% during the shower.

The predictive fan control logic started operating at 00:15 because the system predicted that relative humidity would exceed 70% and water flow was active and

measured relative humidity had already surpassed 60%.

The prediction model demonstrated its ability to forecast relative humidity increases by starting the fan operation before critical thresholds were reached. The proactive system operation enhances comfort while minimizing peak relative humidity levels and enhances overall energy efficiency.

4.4 Limitations and Discussion

Several limitations emerged during the course of this project that are important to acknowledge. First, the dataset used for training and evaluation was collected over a span of only three days, which did not capture any seasonal variations. This limited duration could result in models that are not generalizable across different times of the year or weather conditions. Additionally, the dataset suffered from zero-inflation and a high standard deviation in flow rate measurements (6.27 L/min), which could introduce bias in forecasts, particularly under active shower conditions.

Another limitation concerns the generalizability of the model across different environments and machine learning approaches. TinyML based models exhibited higher mean absolute errors (MAE), which can be attributed to the challenges of training deep networks on small datasets. This suggests that hybrid approaches, such as combining Gradient Boosting with neural networks, could be more effective. Furthermore, the perfect classification accuracy of the fan status raises concerns regarding potential overfitting or the use of rule-based labeling. This highlights the need for greater dataset transparency and validation to ensure that the classification is based on true model inference rather than hard-coded thresholds.

4.5 Conclusion of Analysis

The experimental results confirm that the Gradient Boosting model was the most effective for short-term humidity prediction, achieving reliable accuracy and maintaining low residual errors. The successful deployment and integration of this model into the embedded system demonstrate the feasibility of applying TinyML for real-time environmental control. However, these results also emphasize the importance of careful threshold calibration and the need for larger and more diverse datasets. For future work, addressing temporal variability and enhancing model robustness will be essential steps before considering deployment in industrial or commercial contexts.

The purpose of this study was to develop an intelligent bathroom relative humidity control system using embedded TinyML [9,14]. Specifically, we focused on predicting short-term humidity levels based on water flow and temperature in order to automatically control exhaust fan operation via an ESP32 microcontroller. This chapter critically evaluates the study’s key findings, practical implications, and limitations within the broader context of embedded AI for environmental automation.

5.1 Revisiting Research Questions and Model Findings

The study hypothesized that proactive humidity forecasting through sensor fusion including water temperature, flow rate, and relative humidity would result in more effective fan control than reactive threshold based systems. Our results confirmed this hypothesis. Among the evaluated models, Gradient Boosting achieved the best performance with a Mean Absolute Error (MAE) of 3.31 and an R^2 score of 0.86, demonstrating strong predictive capability and robustness. While this model required slightly longer training, its accuracy and reliability justified the computational cost [21].

These findings highlight that simple ensemble models may outperform neural networks in low-data, resource-constrained environments. However, it is also possible that the relatively poor performance of neural networks (MAE 6.5) resulted from underfitting or suboptimal architecture choices, rather than an inherent flaw in using neural networks for this problem. Additionally, the absence of diverse edge cases in our dataset could have limited model generalizability.

5.2 Analysis of Prediction Quality

Residual analysis revealed a zero-centered, symmetric error distribution, suggesting that the model neither consistently overestimated nor underestimated relative humidity levels. Furthermore, the high correlation coefficient of 0.926 between actual and predicted humidity confirms the model’s ability to generalize within the observed data range. While these metrics are promising, it is important to acknowledge that the limited data collection timeframe may have artificially boosted model performance. The inclusion of broader temporal patterns—such as diurnal or seasonal relative humidity fluctuations—could challenge the model’s robustness and will need to be considered in future iterations.

5.3 Embedded Inference and System Responsiveness

We successfully deployed a quantized TensorFlow Lite model onto the ESP32 microcontroller. This allowed real-time inference every five seconds, enabling the system to predict humidity spikes before they occurred and activate the exhaust fan accordingly [26]. This anticipatory approach resulted in improved responsiveness compared to traditional systems, particularly during high-humidity events such as showers. Figure 4.7 illustrates how predicted humidity levels exceeded the 70% activation threshold ahead of actual values, allowing timely fan operation.

However, the current implementation still relies on static thresholds for fan activation. We believe that incorporating dynamic control mechanisms—such as time-of-day or usage history-based adjustments—could improve energy efficiency and comfort even further. Such enhancements would allow the system to adapt more intelligently to changing user patterns.

5.4 Sensor Fusion and Feature Importance

Among the features used, flow rate emerged as the most influential predictor. Its bimodal distribution accurately reflected typical bathroom activity, alternating between idle and active showering. Although temperature and humidity data provided valuable context, they were less predictive on their own. These findings align with existing research in intelligent buildings, which shows that water use patterns can effectively signal occupancy or ventilation needs [5].

Nevertheless, our system could benefit from additional context-aware features. Incorporating external environmental data such as weather, outdoor humidity, or room occupancy would enhance model adaptability and performance. For a truly comprehensive air quality control solution, integration with CO₂ sensors or VOC (Volatile Organic Compound) detectors may become essential. These additions would support more nuanced environmental monitoring and control.

5.5 Energy Efficiency Assumption and Estimation

The evaluation of predictive fan control effectiveness for energy efficiency required us to estimate the energy usage based on fan ON durations under both reactive and predictive modes. We estimate that the fan power consumption remained constant at 15W throughout the experiment and no hardware power measurement device was deployed. As we assume the fan operated in reactive mode when the measured humidity surpassed 70%. The predictive mode activated the fan when the GradientBoosting model predicted humidity would exceed 70%. The timeseries data points represent 10-second intervals. The average fan runtime was calculated by grouping ON periods into power cycles which resulted in:

The fan operated for 14.2 minutes in reactive mode while it operated for 10.1 minutes in predictive mode.

Using the formula:

$$\text{Energy (Wh)} = \text{Power (W)} \times \text{Time (hours)} \quad (5.1)$$

The Assumed energy consumption was:

$$\text{Reactive: } 15 \text{ W} \times \left(\frac{852 \text{ sec}}{3600} \right) = 3.55 \text{ Wh}$$

$$\text{Predictive: } 15 \text{ W} \times \left(\frac{606 \text{ sec}}{3600} \right) = 2.52 \text{ Wh}$$

$$\begin{aligned} \text{Percentage Saved} &= \left(\frac{\text{Energy}_{\text{Reactive}} - \text{Energy}_{\text{Predictive}}}{\text{Energy}_{\text{Reactive}}} \right) \times 100 \\ &= \left(\frac{3.55 - 2.52}{3.55} \right) \times 100 = 28.9\% \end{aligned}$$

From above calculations we assume an estimated 28.9% reduction in energy consumption per cycle when using prediction-based control.

Note: This estimation assumes ideal power draw and does not factor in real-time current fluctuations or hardware inefficiencies.

5.6 Limitations and Model Generalizability

Despite its success, the system has several limitations that restrict its applicability. First, our dataset was generated from a single bathroom over a short duration, limiting its representation of diverse environments. As a result, the model may not generalize well to different layouts, insulation conditions, or climates.

Second, the feature set was minimal. Important contextual variables such as occupancy, window status, or external weather data were not included. This restricts the model’s capacity to adapt to real-world variations.

Third, we assumed access to a stable power supply. While this was suitable for the prototype, future battery-powered versions will require careful management of inference intervals and energy consumption to maintain portability and efficiency.

Fourth, the design restriction prevents the predictive system from reaching its complete potential. Running the prediction model at all times when water is flowing regardless of current humidity levels would enable more proactive fan activation.

5.7 Practical Implications and Ethical Reflections

The findings from this study suggest that embedded AI can enable practical and affordable solutions for smart home automation. Our system, built with low-cost hardware and minimal sensor requirements, demonstrates that predictive humidity control can improve indoor air quality and energy efficiency simultaneously. This has direct implications for mold prevention and long-term occupant comfort.

Additionally, the system design supports privacy by eliminating the need for cloud connectivity. All processing is done locally, which reduces the risk of data leakage and reliance on external infrastructure [10]. However, the automation of ventilation based on AI decisions introduces ethical considerations. Transparent thresholds, override mechanisms, and user feedback channels are essential for responsible deployment. Users must retain the ability to intervene or understand why certain actions are taken by the system.

5.8 Reconnecting with Research Objectives

The primary research objectives outlined in the introduction were: (1) to explore how embedded ML and sensor fusion could be used to predict humidity levels, (2) to demonstrate real-time predictive control while preserving user privacy, and (3) to evaluate the system’s impact on air quality and energy consumption.

All these objectives were successfully addressed. The predictive model was trained on real data and deployed on hardware, where it demonstrated superior responsiveness compared to static threshold-based systems. The hypothesis that proactive humidity forecasting improves fan control was confirmed by both model evaluation and practical testing. These results support the system’s feasibility as a scalable solution for smart ventilation. However, due to the limited timeframe of this project, we were not able to directly evaluate long-term system efficiency in preventing mold growth, which remains a critical area for future research.

5.9 Placing the Work in a Wider Perspective

This project contributes to the growing field of TinyML by showcasing how TinyML can be leveraged for real-time control in power- and memory-constrained environments. While previous studies have focused on rule-based or cloud-dependent systems, our work emphasizes on-device intelligence, ensuring autonomy, privacy, and rapid responsiveness.

That said, the approach remains context-specific. Bathrooms are ideal for such applications due to their well-defined environmental triggers, but this simplicity also limits the system’s broader applicability. A complete indoor air quality control system would need to monitor additional variables such as CO₂ levels, VOCs, occupancy, and ventilation states.

Other studies have explored hybrid models combining rule-based logic with data-driven insights. We opted for simplicity, focusing on transparency, deployability, and user privacy. Future work could explore combining these approaches for more sophisticated control, especially in multi-room environments.

5.10 Final Reflections

This project began with a simple question: Can we prevent bathroom mold through predictive ventilation using low-cost hardware? The results affirm that we can. By integrating embedded TinyML, sensor fusion, and thoughtful system design, we demonstrated that smart ventilation is achievable without complex infrastructure.

This work highlights that artificial intelligence does not need to be large-scale to be impactful. TinyML opens the door to decentralized, ethical, and responsive environments where decisions are made locally and efficiently. The system we created represents a shift toward intelligent, sustainable home technologies—serving as a meaningful step toward energy-efficient, privacy-conscious, and mold-free living spaces.

6.1 Summary of Contributions

This thesis set out to investigate whether short-term humidity levels in a bathroom environment could be accurately predicted using low-cost sensors and embedded machine learning methods. The broader aim was to enable predictive ventilation control that would reduce mold risk and enhance user comfort, all within the constraints of limited computational resources and a privacy-conscious design.

Through the integration of water flow rate, water temperature, and ambient humidity measurements, we developed a predictive model that achieved a mean absolute error (MAE) of 3.31 and an R^2 score of 0.86. The Gradient Boosting model demonstrated strong predictive capability, even with limited training data. Upon deployment on the ESP32 microcontroller, the system successfully triggered the exhaust fan in advance of reaching critical humidity levels. These results confirm that embedded TinyML systems are capable of delivering intelligent, real-time environmental control without relying on cloud connectivity. Additionally, the developed system demonstrated the ability to reduce energy consumption by up to 29%–30% per shower cycle through shorter fan runtime enabled by predictive control.

6.2 Future Work and Recommendations

While the system performed well in a controlled setting, several directions for future work could enhance its robustness and applicability. A key next step would be to gather datasets from a variety of home environments across different locations and seasons. This would improve the generalizability of the model by capturing a wider range of humidity patterns and use cases.

Expanding the sensor array could also provide richer input data. For instance, incorporating CO₂ and VOC sensors, barometric pressure readings, and occupancy detection mechanisms would allow for more comprehensive environmental understanding. These additional features could help the system better distinguish between ambient conditions and shower-induced humidity spikes.

We also envision the development of user-adaptive features, where threshold values or even the machine learning models themselves could be fine-tuned based on user preferences and behaviors over time. This personalization would enhance comfort and satisfaction while maintaining efficiency.

In terms of optimization, further work could focus on compressing the model through techniques like quantization, pruning, or knowledge distillation. These methods can significantly reduce inference time and memory usage, making the system

even more suitable for ultra-low-power microcontrollers.

Finally, a hybrid approach that leverages cloud-assisted training while maintaining local inference could be explored. This would preserve user privacy while allowing for more complex model updates and learning routines that are otherwise infeasible on-device.

6.3 Conclusion

In conclusion, this project demonstrates that predictive humidity control using embedded TinyML is both technically feasible and practically valuable. The Gradient Boosting model emerged as the most effective solution in a constrained setting, achieving a balance between accuracy and resource efficiency. As real-time responsiveness, local inference, and energy awareness become critical factors in smart home solutions, this work provides a solid foundation for advancing intelligent, privacy-respecting ventilation systems. Our findings represent a step toward creating more autonomous, sustainable, and user-aligned environmental control systems that require no centralized infrastructure yet deliver meaningful performance. The results show that embedded TinyML systems can provide intelligent real-time environmental control without cloud connectivity. The developed system also showed the ability to reduce energy consumption by up to 30% per shower cycle through predictive fan activation.

References

- [1] “Mold allergy-Mold allergy - Symptoms & causes.” [Online]. Available: <https://www.mayoclinic.org/diseases-conditions/mold-allergy/symptoms-causes/syc-20351519>
- [2] O. US EPA, “A Brief Guide to Mold, Moisture and Your Home,” Aug. 2014. [Online]. Available: <https://www.epa.gov/mold/brief-guide-mold-moisture-and-your-home>
- [3] “WHO guidelines for indoor air quality : dampness and mould.” [Online]. Available: <https://www.who.int/publications/i/item/9789289041683>
- [4] N. Gerhardt, “How Much Does Mold Remediation And Removal Cost?” Oct. 2022. [Online]. Available: <https://www.forbes.com/home-improvement/home/mold-remediation-removal-costs/>
- [5] S. Niranjana, R. S. Lavanya, S. Sangeethapriya, N. Nagaraju, B. V. Prasanth, and C. Karthikeyan, “Experimental Procedure to Develop an IoT Based Smart Home Automation System Using Intelligent Sensors Association,” in *2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, Dec. 2024, pp. 1–6. [Online]. Available: <https://ieeexplore.ieee.org/document/10910781>
- [6] Y.-H. Wei, Y.-X. Huang, Y.-C. Chen, and D.-L. Way, “An IoT-based Smart Bathroom for Healthy Life,” in *2023 10th International Conference on Internet of Things: Systems, Management and Security (IOTSMS)*, Oct. 2023, pp. 47–52, iSSN: 2832-3033. [Online]. Available: <https://ieeexplore.ieee.org/document/10325760>
- [7] A. Bind, K. Ashtankar, and M. Chahare, “An iot based smart fan module,” *International Research Journal of Engineering and Technology (IRJET)*, 2020, accessed: 2025-05-26. [Online]. Available: <https://www.irjet.net/archives/V7/i4/IRJET-V7I4238.pdf>
- [8] “IoT Smart Exhaust Fan: ESP32 Based Monitoring & Control.” [Online]. Available: <https://how2electronics.com/iot-smart-exhaust-fan-esp32-based-monitoring-control/>
- [9] “Temperature Prediction using a TinyML LSTM model.” [Online]. Available: <https://www.hackster.io/mjrobot/temperature-prediction-using-a-tinyml-lstm-model-264029>
- [10] M. M. Rahman, M. E. H. Nayeem, M. S. Ahmed, K. A. Tanha, M. S. A. Sakib, K. M. M. Uddin, and H. M. H. Babu, “AirNet: predictive

- machine learning model for air quality forecasting using web interface,” *Environmental Systems Research*, vol. 13, no. 1, pp. 1–19, Dec. 2024. [Online]. Available: <https://environmentalsystemsresearch.springeropen.com/articles/10.1186/s40068-024-00378-z>
- [11] Components101. (2023) YF-S201 Water Flow Measurement Sensor. Accessed: 2025-05-18. [Online]. Available: <https://components101.com/sensors/yf-s201-water-flow-measurement-sensor>
 - [12] Analog Devices. (2023) DS18B20 Digital Thermometer Datasheet. Accessed: 2025-05-18. [Online]. Available: <https://www.analog.com/media/en/technical-documentation/data-sheets/ds18b20.pdf>
 - [13] Aosong Electronics. (2023) DHT22 (AM2302) Temperature and Humidity Sensor Datasheet. Accessed: 2025-05-18. [Online]. Available: <https://cdn.sparkfun.com/assets/f/7/d/9/c/DHT22.pdf>
 - [14] “Water Leakage detection system with tinyML framework.” [Online]. Available: <https://www.embien.com/case-studies/water-leakage-detection-system-with-tinymml-framework>
 - [15] “Pipeline Clog Detection with a Flowmeter and TinyML.” [Online]. Available: <https://www.hackster.io/coderscafe/pipeline-clog-detection-with-a-flowmeter-and-tinymml-86155d>
 - [16] “(PDF) IoT Based Detection of Molded Bread and Expiry Prediction using Machine Learning Techniques,” Feb. 2022. [Online]. Available: https://www.researchgate.net/publication/363052562_IoT_Based_Detection_of_Molded_Bread_and_Expiry_Prediction_using_Machine_Learning_Techniques
 - [17] “Indoor Air Quality Monitoring and Controlling System based on IoT and Fuzzy Logic.” [Online]. Available: https://ieeexplore.ieee.org/abstract/document/8835246?casa_token=zI1l75yQM_sAAAAA:jKP7EaGNk46nAFO1l1qKoAzWd_gwhGBkrLH5P_rU8m0RPG2Ody0F_A46LlqfiPxrqaGGmRODGw
 - [18] “A Method for Forecasting Indoor Relative Humidity for Improving Comfort Conditions and Quality of Life.” [Online]. Available: <https://www.mdpi.com/2073-4433/16/3/315>
 - [19] “An autoML algorithm to select linear regression model and its performance evaluation for linear datasets.” [Online]. Available: <https://ieeexplore.ieee.org/document/10924008>
 - [20] L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, Oct. 2001. [Online]. Available: <https://link.springer.com/article/10.1023/A:1010933404324>
 - [21] “(PDF) IoT Based Detection of Molded Bread and Expiry Prediction using Machine Learning Techniques.” [Online]. Available: https://www.researchgate.net/publication/363052562_IoT_Based_Detection_of_Molded_Bread_and_Expiry_Prediction_using_Machine_Learning_Techniques

- [22] M. Awad and R. Khanna, “Support Vector Regression,” in *Efficient Learning Machines*. Apress, Berkeley, CA, 2015, pp. 67–80. [Online]. Available: https://link.springer.com/chapter/10.1007/978-1-4302-5990-9_4
- [23] Espressif Systems. (2023) ESP32 Series Datasheet. Accessed: 2025-05-18. [Online]. Available: https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf
- [24] P. Schober, C. Boer, and L. A. Schwarte, “Correlation Coefficients: Appropriate Use and Interpretation,” *Anesthesia & Analgesia*, vol. 126, no. 5, p. 1763, May 2018. [Online]. Available: https://journals.lww.com/anesthesia-analgesia/fulltext/2018/05000/Correlation_CoefficientsAppropriate_Use_and.50.aspx
- [25] “Pearson Correlation Coefficient | SpringerLink.” [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-642-00296-0_5
- [26] Y. Liu, Z. Yang, Y. Zhang, H. Zhang, and Q. Wei, “A review on tinymt: State-of-the-art and prospects,” *Journal of King Saud University - Computer and Information Sciences*, vol. 34, no. 8, pp. 6654–6669, 2022, accessed: 2025-05-30. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1319157821003335>

